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DNA recombination through assembly graphs

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ABSTRACT

Motivated by DNA rearrangements and DNA homologous recombination modeled in [A. Angeleska, N. Jonoska, M. Saito, L.F. Landweber, RNA-guided DNA assembly, Journal of Theoretical Biology, 248(4) (2007), 706–720], we investigate smoothings on graphs that consist of only 4-valent and 1-valent rigid vertices, called assembly graphs. An assembly graph can be seen as a representation of the DNA during certain recombination processes in which 4-valent vertices correspond to the alignment of the recombination sites. A single gene is modeled by a polygonal path in an assembly graph. A polygonal path makes a "right-angle" turn at every vertex, defining smoothings at the 4-valent vertices and therefore modeling the recombination process. We investigate the minimal number of polygonal paths visiting all vertices of a given graph exactly once, and show that for every positive integer n there are graphs that require at least n such polygonal paths. We show that there is an embedding in three-dimensional space of each assembly graph such that smoothing of vertices according to a given set of polygonal paths results in an unlinked graph. As some recombination processes may happen simultaneously, we characterize the subsets of vertices whose simultaneous smoothings keep a given gene in tact and give a characterization of all sequences of sets of vertices defining successive simultaneous smoothings that can realize complete gene rearrangement.

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1. Introduction

Massive DNA rearrangements and DNA recombination have been observed in some species of ciliates, and they are often taken as model systems for studies of DNA rearrangements that may occur more generally [9]. Theoretical investigations of DNA rearrangements in ciliates through string-rewriting operations and graph reduction operations can be found in [3], and more recently in [2]. In this paper, we propose combinatorial models for a variety of problems that arise in these recombination processes, using spatial graphs with 4-valent vertices, paths in graphs, and smoothing of vertices. Some parts of the biological motivation and background are described in Section 2.

We introduce a notion of an assembly graph[1] to model the DNA structure during the recombination process, and study the vertex removal in such graphs (called smoothing) that model the resulting molecules after recombination. A consecutive sequence of edges representing a possible smoothing in the assembly graph is called a "polygonal path". The authors of [4] used an operation "folding" on cyclic graphs to obtain a graph structure analogous to our notion of assembly graphs and an operation "unfolding" similar to our notion of smoothing. The main result in [4] shows that every micronuclear gene sequence can be represented as a "folded graph" such that the resulting macronuclear gene is obtained as a set of "unfoldings". Similarly, in this paper, each polygonal path in an assembly graph models a resulting macronuclear gene obtained after recombination. Recent studies show that some gene segments of a single gene can appear on more than

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[1] The term "assembly graph" is also used in some models used for DNA sequencing in a different context.

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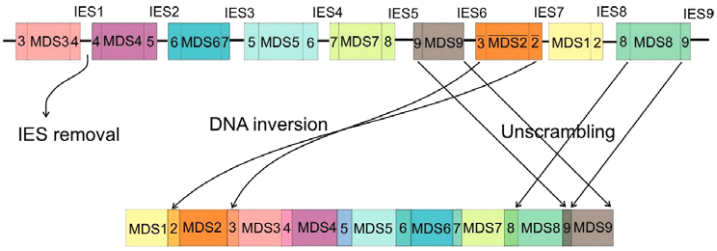

Fig. 1. Schematic representation of the scrambled Actin I micronuclear germline gene in *Oxytricha nova* (top) and the correctly assembled macronuclear gene (bottom). Each block represents an MDS, and each line between blocks is an IES. The numbers at the beginning and at the end of each segment represent the pointer sequences. Note that MDS3· · ·MDS8 require permutation and inversion to assemble into the orthodox, linear order MDS1· · ·MDS9 in the macronucleus. The bars above MDS2 and its pointers indicate that this block is inverted relative to the others, i.e., this sequence is the Watson–Crick reverse complement of the version in the macronucleus; from [12].

one locus [8] (see also [9]). Therefore, we study the number of genes that can result after a given set of rearrangements, which is modeled by the number of polygonal paths in a given assembly graph. A set of polygonal paths that visits every vertex in the assembly graph once is called here a "Hamiltonian set". We show that in general there are no restrictions on the minimal number of polygonal paths that may appear in assembly graphs. Other properties showing relationship between the number of vertices in the assembly graph and the number of minimal polygonal paths are obtained in Section 4.

In Section 5 we study smoothing of vertices and the resulting structures. In order to have a biologically functional gene, it is desirable that, after all necessary rearrangements are performed, the resulting molecule is unknotted. We show that, given an assembly graph and a Hamiltonian set of polygonal paths, there is always an embedding of the graph in $\mathbb{R}^3$ such that the smoothing of the graph results in unlinked components. Further, we compare the smoothings defined by polygonal paths with existent smoothings of virtual knot diagrams defined through proper coloring introduced by Kauffman [7]. We show that, in general, these two types of smoothing on assembly graphs (virtual knot diagrams) are not equal. Therefore, smoothings defined by polygonal paths may be an interesting new concept for investigation mathematically.

In [1] (as also follows from the result in [4]) it was shown that, in an assembly graph with a single Hamiltonian polygonal path, the simultaneous smoothing of all vertices always results with one component containing all gene segments in the proper functional order of the gene. Although it is natural to expect that some of these rearrangements occur simultaneously, or in a close time frame difference to be considered as simultaneous, it is probably not the case that all of them occur at once. The last section of our paper is concerned with partial and successive simultaneous rearrangements, that is, identifying the sets of vertices that can be smoothed simultaneously without disconnecting the polygonal paths. Parallel recombination based on the string-rewriting operations and corresponding graph reductions were also investigated in [5]. Here we give necessary and sufficient conditions for a set of vertices (hence set of pointers) in assembly graphs to allow "successful" simultaneous smoothings that keep polygonal paths connected, and therefore all of the gene segments on a single molecule. We also characterize all possible pathways of rearrangements for a complete successful assembly of a given gene.

2. Biological motivations and background

A model for homologous DNA recombination events that are guided by either double stranded RNA (dsRNA) or single stranded RNA (ssRNA) templates was presented in [1]. This model takes as a case study the DNA rearrangements that take place in some groups of ciliates, such as *Stylonychia* or *Oxytricha*, where extensive gene rearrangement while differentiating somatic macronuclei from germline micronuclei takes place. The resulting "nanochromosomes" of the macronucleus in spirotrichous ciliates typically encode a single gene. Each gene in the macronucleus may be present in the micronucleus as several non-consecutive segments (macronuclear destined sequences, MDSs) separated by non-coding DNA. During macronuclear differentiation, the non-coding fragments (internal eliminated sequences, IESs) that interrupt MDSs in the micronucleus are deleted. Moreover, the order of the MDSs in the micronucleus may not be consecutive, and some of them might be reversed relative to the others. Therefore, formation of the macronucleus requires unscrambling of the MDS order, MDS inversion, as well as IES removal (see Fig. 1). The DNA rearrangements in ciliates are believed to be enabled by pointer sequences (pairs of identical sequences of nucleotides) that assure correct splicing position. A model based on RNA-guided DNA rearrangements was proposed in [1]. Recently, this model was supported by an experimental observation that maternal RNA templates may guide the DNA rearrangement in the early development of the macronucleus [11].

In the model presented in [1], the recombination can be seen as a 4-valent vertex made of two DNA segments. The model utilizes graphs as a physical representation of the DNA at the time of recombination. Schematically, the moment of the homologous recombination process can be represented as a vertex in a graph as depicted in Fig. 2(left). The homologous recombination corresponds to removal of a crossing in the knot diagram called "smoothing" (see Fig. 2(right)). The RNA or DNA template enables alignment of pointers and, as shown in [1], this alignment (being parallel or antiparallel) determines the type of recombination and the type of smoothing performed at a vertex.

Also, in [1] it is observed that, given a micronuclear DNA sequence containing a single rearranged macronuclear gene, if all recombination events are performed simultaneously, one of the resulting molecules always contains all of the MDSs

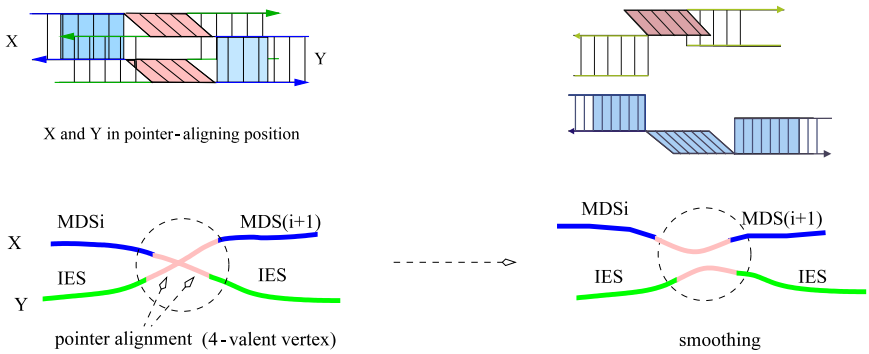

Fig. 2. **(Left)** Schematic representation of the pointer alignment shown as a 4-valent vertex. Two micronuclear DNA segments X and Y (double stranded molecules schematically represented with ladders) exchange pointer nucleotides are indicated as pale (pink) through branch migration (top) with MDSs segments indicated as bold (blue) (see [1] for details). This alignment is represented as a 4-valent vertex in a graph (bottom). **(Right)** Two molecules after the recombination (top), with the MDSs joined on the same molecule. Schematic representation of the finished recombination as smoothing of the vertex (bottom). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

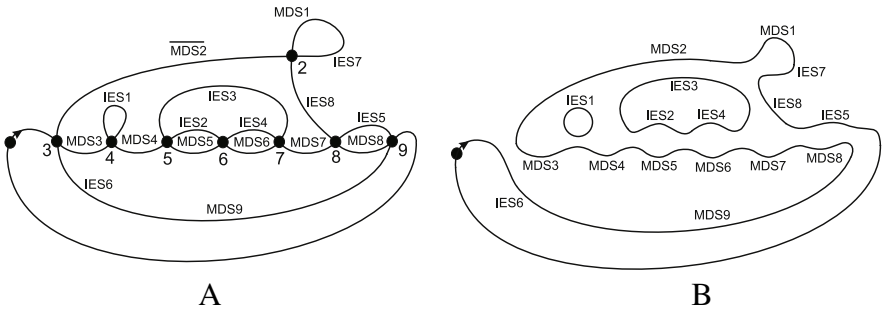

Fig. 3. Schematic representation of simultaneous recombination, at the moment of recombination (left), and resulting molecules after recombination (right).

in a linear macronuclear gene order. The graph depicted in Fig. 3(A) is a schematic planar representation of the MDS-IES structure of the scrambled gene in Fig. 1 at the time of rearrangement. The resulting graph, after smoothing of every vertex, is depicted in Fig. 3(B).

Here we investigate the process of DNA rearrangements combinatorially, by formalizing the MDS-IES structures by graphs with rigid 4-valent vertices. Each rigid vertex corresponds to a pair of pointers aligned at the time of recombination. The edges correspond to either MDS or IES sequences. As in [1], the DNA recombination events are formalized through smoothings of the rigid vertices.

3. Preliminaries

We start with defining rigid vertices. Although graphs with rigid vertices commonly appear in knot theory (see, for example, [6]), we present the definitions for self-containment of the exposition and establish notations. For an n-tuple $\mathbf{x} = (x_1, \ldots, x_n)$ of elements of a set X, we define its *reverse* to be $(x_n, \ldots, x_1)$ which is denoted by $\mathbf{x}^R = (x_1, \ldots, x_n)^R$. Also define the set $\mathbf{x}^{\text{rev}} = (x_1, \ldots, x_n)^{\text{rev}} = \{\mathbf{x}, \mathbf{x}^R\}$. Two n-tuples $\mathbf{x}$ and $\mathbf{y}$ are called *reverse equivalent* if $\mathbf{x}^{\text{rev}} = \mathbf{y}^{\text{rev}}$, and denoted by $\mathbf{x} \sim_{\text{rev}} \mathbf{y}$. Given an n-tuple $\mathbf{x} = (x_1, \ldots, x_n)$, we denote by $\mathbf{x}^{\text{cyc}} = (x_1, \ldots, x_k)^{\text{cyc}}$ the set:

$$(x_1, \ldots, x_k)^{\text{cyc}} = \{(x_1, \ldots, x_k), (x_2, \ldots, x_k, x_1), \ldots, (x_k, x_1, \ldots, x_{k-1}), (x_k, \ldots, x_1), (x_{k-1}, \ldots, x_1, x_k), \ldots, (x_1, x_k \ldots, x_2)\}$$

which consists of all compositions of cyclic permutations and reverses of $(x_1, \ldots, x_k)$. Two n-tuples $\mathbf{x}$ and $\mathbf{y}$ are called *cyclically equivalent* if $\mathbf{x}^{\text{cyc}} = \mathbf{y}^{\text{cyc}}$, which is denoted by $\mathbf{x} \sim_{\text{cyc}} \mathbf{y}$. These notions are defined in the same manner for multisets as well.

Let $\Gamma = (V, E)$ be a finite graph with a set of vertices V and a set of edges E. We consider, topologically, Γ as a 1-dimensional complex with V regarded as 0-simplices and E as 1-simplices [10]. We allow multiple edges and loops. Denote by $E(v)$ the multiset of edges that are incident to a vertex $v \in V$. An edge in $E(v)$ is repeated if and only if it is a loop based at v. The multiset cardinality of $E(v)$ is called the *valency of* v. Let $v \in V$ and $E(v) = \{e_1, \ldots, e_k\}$. For each v we fix an order of $E(v)$ written by $(e_{i_1}, \ldots, e_{i_k})$. We denote by $E^{\text{cyc}}(v)$ the set $(e_{i_1}, \ldots, e_{i_k})^{\text{cyc}}$. Diagrammatically, the vertex v can be considered as a small disk such that incident edges are attached to the boundary of the disk at a point called the "entering point" of e.

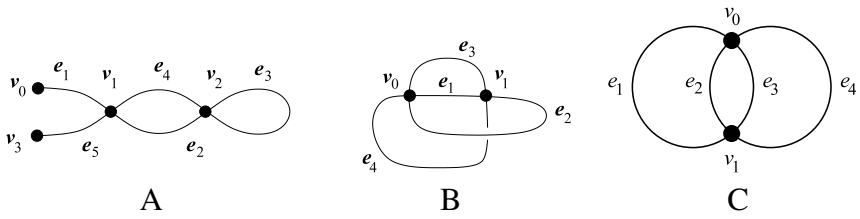

Fig. 4. Examples of assembly graphs. Simple assembly graphs (A) and (B), and a non-simple assembly graph (C).

The edges incident to v are sketched such that if one traces the boundary of the disk clockwise, the entering points of the edges from $E(v)$ are encountered in the order found in $E^{\text{cyc}}(v)$. In Fig. 4, such small disks are represented by small black dots. In Fig. 4 (B), there is a crossing with the "under-arc" broken as commonly seen in knot theory, which is not a vertex of the graph. The graph is depicted as a projection of a spatial graph with crossing information provided. A *rigid vertex* in a finite graph $\Gamma = (V, E)$ is a pair $(v, E^{\text{cyc}}(v))$. If $(v, (e_1, \ldots, e_k)^{\text{cyc}})$ is a rigid vertex then the edges e_{i-1} and e_{i+1} are called *neighbors* of e_i, for $i = 2, \ldots, k-1$; the neighbors of e_k are e_{k-1} and e_1; the neighbors of e_1 are e_k and e_2. All vertices in this paper are rigid vertices; therefore in what follows we use only the name of the vertex to indicate a rigid vertex, understanding that the order of the edges is predetermined.

Definition 3.1. An *assembly graph* is a finite connected graph, where all vertices are rigid vertices of valency 1 or 4. A vertex of valency 1 is called an *endpoint*.

The number of 4-valent vertices in Γ is denoted with $|\Gamma|$. The assembly graph is called *trivial* if $|\Gamma| = 0$. Note that the definition of an assembly graph implies that the number of endpoints is always even.

Definition 3.2. Two assembly graphs $\Gamma_1 = (V_1, E_1)$ and $\Gamma_2 = (V_2, E_2)$ are *isomorphic* if there is a graph isomorphism Φ that preserves the cyclic order of each rigid vertex. More specifically, for a graph isomorphism $\Phi = (\Phi_v, \Phi_e : \Gamma_1 \to \Gamma_2)$ with $\Phi_v : V_1 \to V_2$ and $\Phi_e : E_1 \to E_2$, for every rigid vertex $(v, (e_1, e_2, e_3, e_4)^{\text{cyc}})$ in Γ_1, we have $(\Phi_v(v), (\Phi_e(e_1), \Phi_e(e_2), \Phi_e(e_3), \Phi_e(e_4))^{\text{cyc}}) = (\Phi_v(v), E^{\text{cyc}}(\Phi_v(v)))$.

As is often the case, we may abuse the notation for assembly graph isomorphism by writing Φ instead of either Φ_v or Φ_e whenever it is clear from the context which map is used.

Definition 3.3. A *transverse path* in Γ is a sequence $\gamma = (v_0, e_1, v_1, e_2, \ldots, e_n, v_n)$ if v_0, v_n are endpoints, or $(v_0, e_1, v_1, e_2, \ldots, e_n)$, if v_0 is a 4-valent vertex and $e_n \in E(v_0)$, satisfying the following conditions: (1) $v_0, \ldots, v_n$ is a sequence of a subset of vertices of Γ, with possible repetition of the same vertex at most twice, (2) $\{e_1, \ldots, e_n\}$ is a set of distinct edges, and (3) each e_i is not a neighbor of e_{i-1} with respect to the rigid vertex v_{i-1}, $i = 2, \ldots, n$, and in the case where v_0 is a 4-valent vertex, e_1 is not a neighbor of e_n with respect to the rigid vertex v_0.

A transverse path can be considered as an image of a map from the unit interval $[0, 1]$ to Γ, where the image of the boundary points $(\{0\} \cup \{1\})$ consists of either two endpoints of Γ, or a single 4-valent vertex.

In Fig. 4 examples of assembly graphs are depicted. Note that the graphs in (B) and (C) do not have any endpoints, and the broken under-arc depicts the crossing information as in standard knot diagrams, as already mentioned. A transverse path for each example is given by

$$\text{(A) } (v_0, e_1, v_1, e_2, v_2, e_3, v_2, e_4, v_1, e_5, v_3), \qquad \text{(B) } (v_0, e_1, v_1, e_2, v_0, e_3, v_1, e_4), \quad \text{and} \quad \text{(C) } (v_0, e_2, v_1, e_4).$$

Intuitively, transverse paths trace the cyclic or linear *transverse components* of an assembly graph, so the number of transverse paths is equal to the number of transverse components in the graph. In the case of (C), there is another transverse path (v_0, e_3, v_1, e_1), which is not equivalent to (v_0, e_2, v_1, e_4) in the following sense.

Definition 3.4. Two transverse paths with endpoints are *equivalent* if they are either identical or one is the reverse of the other. Two transverse paths γ, γ' without endpoints are *equivalent* if they have the same cyclic order: $\gamma^{\text{cyc}} = \gamma'^{\text{cyc}}$.

Definition 3.5. An assembly graph Γ is called *simple* if there is a transverse *Eulerian* path in Γ, meaning, there is a transverse path γ that contains every edge from Γ exactly once.

In Fig. 4, the graphs (A) and (B) are simple, but the graph (C) is not. Note that in a simple assembly graph, a Eulerian transverse path visits every non-endpoint vertex twice.

Lemma 3.6. *In a simple assembly graph, there is a unique equivalence class of transverse Eulerian paths.*

Proof. By the definition of transverse paths, a Eulerian path γ in a simple assembly graph Γ either starts and ends with two endpoints, or none of the vertices in γ are endpoints. If γ has two endpoints, then every Eulerian path in Γ has to start and

end at the endpoints. Starting from an endpoint, following its incident edge, at every consecutive vertex there is only one edge that is not a neighbor of the edge just passed. Hence there is only one transverse Eulerian path starting at one endpoint. Since the reverse of this path is a transverse Eulerian path from the other endpoint, the two transverse Eulerian paths are equivalent.

Now consider the case when Γ and therefore γ has no endpoints. We will show that every Eulerian path in Γ is equivalent to γ, meaning, it belongs to γ^{cyc}. Let $\gamma = (v_0, e_1, v_1, e_2, \ldots, e_n)$. Since none of the vertices are endpoints, all of them are 4-valent. Therefore, each vertex v appears in γ exactly twice. Denote the positions of v in γ by i and j $(i < j)$, i.e., $v_i = v_j = v$. Let $\alpha_1 = v_0, e_1, \ldots, e_i$, $\alpha_2 = e_{i+1}, \ldots, e_j$, $\alpha_3 = e_{j+1}, \ldots, e_n$. In the case of $i = 0$, α_1 is empty. Then γ can be rewritten as $\gamma = (\alpha_1, v_i, \alpha_2, v_j, \alpha_3)$. There are four possibilities for a Eulerian path to start at $v = v_i = v_j$ choosing either one of the edges $e_i, e_{i+1}, e_j, e_{j+1}$. Note that in either one of these choices the successive edges are uniquely determined, since at each vertex there is only one non-neighboring edge to the edge just passed. Hence the choice of the first edge uniquely determines the Eulerian path. Therefore there are only four Eulerian paths that start at v: $(v_i, \alpha_2, v_j, \alpha_3, \alpha_1)$, $(v_i, \alpha_1^R, \alpha_3^R, v_j, \alpha_2^R)$, $(v_j, \alpha_3, \alpha_1, v_i, \alpha_2)$, and $(v_j, \alpha_2^R, v_i, \alpha_1^R, \alpha_3^R)$, and all four of these paths belong to γ^{cyc}. □

The following lemma shows that the simple assembly graphs are uniquely determined (up to isomorphism) by the transverse Eulerian path.

Lemma 3.7. *Let $\Gamma = (V, E)$ and $\Gamma' = (V', E')$ be two simple assembly graphs with γ and γ' their transverse Eulerian paths, respectively. Then Γ and Γ' are isomorphic if and only if there is a map $\Phi = (\Phi_v, \Phi_e) : \Gamma \to \Gamma'$ with bijections $\Phi_v : V \to V'$ and $\Phi_e : E \to E'$ such that $\Phi(\gamma)$ is equivalent to γ'.*

Proof. If Γ and Γ' are isomorphic by an isomorphism Φ, then $\Phi(\gamma)$ is a transverse Eulerian path in Γ', and by Lemma 3.6, γ' and $\Phi(\gamma)$ are equivalent. Conversely, the pair of bijections $\Phi = (\Phi_v, \Phi_e)$ provides a graph isomorphism as the transverse Eulerian path γ maps to a transverse Eulerian path $\Phi(\gamma)$. Hence it suffices to show that the rigidity of the vertices is preserved: $(\Phi_v(v), (\Phi_e(e_1), \Phi_e(e_2), \Phi_e(e_3), \Phi_e(e_4))^{\text{cyc}}) = (\Phi_v(v), E^{\text{cyc}}(\Phi_v(v)))$ for every vertex v in Γ. Let v be a 4-valent vertex in Γ, and let $\gamma = (v_0, e_1, v_1, e_2, \ldots, e_n, (v_n))$. Then there are $i, k \in \{1, 2, \ldots, n\}$ such that $v = v_i = v_k$. Then $\Phi_v(v) = \Phi_v(v_i) = \Phi_v(v_k)$ and we have $E^{\text{cyc}}(v) = (e_{i-1}, e_{k-1}, e_i, e_k)^{\text{cyc}}$ and $E^{\text{cyc}}(\Phi_v(v)) = (\Phi_e(e_{i-1}), \Phi_e(e_{k-1}), \Phi_e(e_i), \Phi_e(e_k))^{\text{cyc}}$; therefore $(\Phi_v(v), (\Phi_e(e_1), \Phi_e(e_2), \Phi_e(e_3), \Phi_e(e_4))^{\text{cyc}}) = (\Phi_v(v), E^{\text{cyc}}(\Phi_v(v)))$ holds. □

Note: Further in this section, unless otherwise stated, all assembly graphs are simple assembly graphs.

We establish a convention of representing simple assembly graphs by words. A finite set Σ is called an *alphabet* and a *word* is an element in the free monoid Σ^*. The number of occurrences of a symbol s in a word w is denoted $|w|_s$. For $w \in \Sigma^*$ the *domain* of w is the set $\text{dom}(w) = \{s \in \Sigma : |w|_s \geq 1\}$. A *double occurrence word* over alphabet Σ is a word w such that, for every $s \in \Sigma$, $s \in \text{dom}(w)$ implies $|w|_s = 2$. The *length* $|w|$ of a word w is the cardinality of $\text{dom}(w)$. Two words u and v are *equivalent* if there is a bijection $g : \text{dom}(u) \to \text{dom}(v)$ that induces the monoid homomorphism $g_* : \text{dom}(u)^* \to \text{dom}(v)^*$ such that $g_*(u) = w$.

For a word $w = w_1 w_2 \cdots w_n$, we continue to use the notation w^{cyc} denoting $(w_1, w_2, \ldots, w_n)^{\text{cyc}}$, and w^{rev} for $\{w, w^R\}$, where $w^R = w_n \cdots w_2 w_1$. We say that v is *reverse equivalent* to w ($v \sim_{\text{rev}} w$) if $v^{\text{rev}} = w^{\text{rev}}$, and v is *cyclic equivalent* ($v \sim_{\text{cyc}} w$) to w if $v^{\text{cyc}} = w^{\text{cyc}}$. In the rest of the exposition we always work with an equivalence class representative of either ^{rev} or ^{cyc}.

Lemma 3.8. (a) *There is a one-to-one correspondence between the isomorphism classes of simple assembly graphs with two endpoints and the reverse equivalence class of double occurrence words.* (b) *There is a one-to-one correspondence between the isomorphism classes of simple assembly graphs with no endpoints and the cyclic equivalence classes of double occurrence words.*

The above Lemma 3.8 is well known in knot theory as a method of representing knot diagrams by "Gauss codes" (see, for example, [6]). Here we briefly describe the bijective correspondence. Let Γ be a simple assembly graph with two endpoints, and let $\gamma = (v_0, e_1, v_1, e_2, \ldots, e_n, v_n)$ be a transverse Eulerian path in Γ. Then the corresponding double occurrence word is defined to be $w = v_1 \cdots v_{n-1}$, where the vertices are regarded as a set of symbols. In the case of a simple assembly graph with no endpoints and with a transverse Eulerian path $\gamma = (v_0, e_1, \ldots, e_n)$, the corresponding double occurrence word is $v_0 v_1 \cdots v_{n-1}$. Conversely, for a given double occurrence word $a_1 \cdots a_n$ over some alphabet Σ, we can construct a simple assembly graph by connecting vertices labeled by $a_1, \ldots, a_n$ consecutively from a_1 to a_2, a_2 to a_3, and so on, in a transverse manner, i.e., at every vertex the outgoing edge is not a neighbor of the incoming edge. Similarly, one can obtain an assembly graph with no endpoints by adding an edge from a_n to a_1.

Example 3.9. There are three reverse equivalences for double occurrence words in alphabet $\Sigma = \{a, b\}$ that correspond to three equivalence classes of simple assembly graphs with two endpoints, represented by double occurrence words *aabb*, *abab* and *abba*. The assembly graph in Fig. 4 (A) corresponds to the double occurrence word *abba* by mapping $v_1 \mapsto a$, $v_2 \mapsto b$.

There are only two cyclic equivalences for double occurrence words in alphabet $\Sigma = \{a, b\}$ that correspond to two equivalence classes of simple assembly graphs with no endpoints. They are represented by the words *aabb* and *abab*.

Given a simple assembly graph Γ with two endpoints, choose one of them to be initial (i) and the other to be terminal (t). We call such Γ a *directed* simple assembly graph with direction from i to t. We consider the transverse path of a directed simple assembly graph as a path starting at the vertex i and terminating at the vertex t.

Definition 3.10. A *composition* $\Gamma_1 \circ \Gamma_2$ of two (directed simple) assembly graphs Γ_1 and Γ_2 is the directed simple assembly graph, obtained by identifying the terminal vertex of Γ_1 with the initial vertex of Γ_2.

Note that the initial vertex of $\Gamma_1 \circ \Gamma_2$ is the initial vertex of Γ_1 and the terminal vertex of $\Gamma_1 \circ \Gamma_2$ is the terminal vertex of Γ_2. If the double occurrence words u and v correspond to the directed simple assembly graphs Γ_1 and Γ_2, respectively, then the concatenation uv corresponds to the composition $\Gamma_1 \circ \Gamma_2$. In general, $\Gamma_1 \circ \Gamma_2$ is not isomorphic to $\Gamma_2 \circ \Gamma_1$; for example take Γ_1 represented by *aa* and Γ_2 by *bbcddc*.

Definition 3.11. If a double occurrence word w can be written as a product $w = uv$ of two non-empty double occurrence words u, v, then w is called *reducible*; otherwise it is called *irreducible*. Similarly, if $\Gamma = \Gamma_1 \circ \Gamma_2$ for some non-trivial directed assembly graphs Γ_1 and Γ_2, then Γ is called *reducible*. Otherwise it is called *irreducible*.

Note that if w corresponds to an assembly graph Γ with no endpoints, then every $v \in w^{\text{cyc}}$ is also a representation of Γ, and therefore the notion of "reducible" words and/or graphs does not apply. For example, $abba \sim_{\text{cyc}} bbaa$.

4. Polygonal paths and assembly numbers

In this section we define polygonal paths in assembly graphs as a model for a single macronuclear gene. The edges of a polygonal path represent MDSs such that neighboring edges represent successive MDS sequences. The vertices where the edges meet model the pointer sequences which guide the assembly of the gene. Recent studies show that a micronuclear DNA sequence may contain scrambled MDS portions of several genes, or even a single gene may appear on different loci [8,9]. Thus we consider the question: Given a micronuclear DNA sequence, and a set of pointers indicating the necessary rearrangements, what is the minimal number of macronuclear genes that such a sequence can reproduce? This question is modeled through investigating the structure of a single micronuclear molecule (from the viewpoint of pointer sequence arrangements) and the number of genes that can be obtained through homologous recombination. A set of polygonal paths which cover all vertices in an assembly graph, therefore, models the set of corresponding macronuclear genes obtained as a result of the recombination. The assembly number for the assembly graph investigated in this section corresponds to the minimal number of genes that can result from the DNA rearrangements.

Definition 4.1. Let Γ be an assembly graph. An *open path* in Γ is a homeomorphic image of the open interval $(0, 1)$ in Γ.

An open path is also represented by a sequence:

$$((e_1 \setminus v_0), v_1, e_2, v_2, e_3, \ldots, v_{m-1}, e_m, v_m, (e_{m+1} \setminus v_{m+1})),$$

where the v_i's are vertices in Γ for $i \in \{1, 2, \ldots, m\}$ such that $v_i \neq v_j$ when $i \neq j$, and the e_i's are edges in Γ for $i \in \{1, 2, \ldots, m\}$ with endpoints v_{i-1} and v_i respectively, such that the initial vertex of e_1 (and possibly part of e_1) and the terminal vertex of e_{m+1} (and possibly part of e_{m+1}) are not included. We say that the open path is a *cycle* if $e_1 = e_{m+1}$.

Two open paths are *disjoint* if they do not have a vertex in common.

Definition 4.2. A set of pairwise disjoint open paths $\{\gamma_1, \ldots, \gamma_k\}$ in Γ is called *Hamiltonian* if their union contains all 4-valent vertices of Γ. An open path γ is called *Hamiltonian* if the set $\{\gamma\}$ is Hamiltonian.

Definition 4.3. A *polygonal* path is an open path γ: $((e_1 \setminus v_0), v_1, e_2, \ldots, v_{m-1}, e_m, v_m, (e_{m+1} \setminus v_{m+1}))$, such that e_i and e_{i+1} are neighbors for every $i \in \{1, 2, \ldots, m\}$.

Intuitively, the polygonal paths indicate the way a DNA rearrangement can occur, and indicate the types of smoothings that correspond to these rearrangements.

Definition 4.4. Let Γ be an assembly graph. The *assembly number* of Γ, denoted by $\text{An}(\Gamma)$, is defined by $\text{An}(\Gamma) = \min\{\, k \mid$ there exists a Hamiltonian set of polygonal paths $\{\gamma_1, \ldots, \gamma_k\}$ in $\Gamma\}$.

Motivated by realizable words discussed in [3], a simple assembly graph Γ is called *realizable* if $\text{An}(\Gamma) = 1$; otherwise it is called *unrealizable*.

Example 4.5. The assembly graph depicted in Fig. 4 (A) has a Hamiltonian path represented by $(e_1 \setminus v_0), v_1, e_2, v_2, (e_3 \setminus v_2)$ that is not polygonal. In Fig. 5, polygonal Hamiltonian paths are depicted by thick dotted curves. Note that for the graph Γ depicted in (B), there is no single component polygonal Hamiltonian path, and therefore $\text{An}(\Gamma) = 2$, and Γ is unrealizable.

Lemma 4.6. *For each pair of directed simple assembly graphs Γ_1 and Γ_2, one of the following equalities hold:* $\text{An}(\Gamma_1 \circ \Gamma_2) = \text{An}(\Gamma_1) + \text{An}(\Gamma_2)$, *or* $\text{An}(\Gamma_1 \circ \Gamma_2) = \text{An}(\Gamma_1) + \text{An}(\Gamma_2) - 1$.

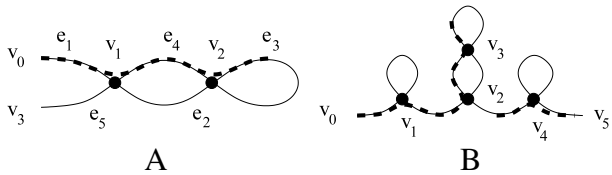

Fig. 5. Polygonal Hamiltonian paths.

Proof. Let $\mathrm{An}(\Gamma_1) = k_1$ and $\mathrm{An}(\Gamma_2) = k_2$. There are Hamiltonian sets of polygonal paths $S_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_{k_1}\}$ for Γ_1 and $S_2 = \{\beta_1, \beta_2, \ldots, \beta_{k_2}\}$ for Γ_2, respectively. Then $S_1 \cup S_2 = \{\alpha_1, \alpha_2, \ldots, \alpha_{k_1}, \beta_1, \beta_2, \ldots, \beta_{k_2}\}$ is a Hamiltonian set of polygonal paths for $\Gamma_1 \circ \Gamma_2$. Hence

$$\mathrm{An}(\Gamma_1 \circ \Gamma_2) = k \le k_1 + k_2 = \mathrm{An}(\Gamma_1) + \mathrm{An}(\Gamma_2).$$

Let S be a Hamiltonian set of polygonal paths for $\Gamma_1 \circ \Gamma_2$ with minimal number of components (i.e., S consists of k distinct paths). As $\Gamma_1 \circ \Gamma_2$ is obtained by identifying one endpoint from Γ_1 and one endpoint of Γ_2, S contains at most one path that includes vertices from both Γ_1 and Γ_2. If S does not contain a path that includes vertices from both Γ_1 and Γ_2, then there are at least k_1 paths of S that contain all vertices of $\Gamma_1 \subset \Gamma_1 \circ \Gamma_2$, and at least k_2 paths for $\Gamma_2 \subset \Gamma_1 \circ \Gamma_2$, so that $k \ge k_1 + k_2$, and we obtain $k = k_1 + k_2$. Suppose S contains a path γ that includes vertices both from Γ_1 and Γ_2. Let $\gamma = \gamma_1\gamma_2$, where γ_i contains vertices from Γ_i only, and let $S_i \subset S$ be the paths that visit vertices from Γ_i only ($i = 1, 2$). Each S_i contains at least $k_i - 1$ components since $S_i \cup \{\gamma_i\}$ is a Hamiltonian set of polygonal paths for Γ_i. Hence we obtain $k \ge (k_1 - 1) + (k_2 - 1) + 1$. Therefore, in this case, $k = k_1 + k_2 - 1$. □

Remark 4.7. We note that $\mathrm{An}(\Gamma_1 \circ \Gamma_2) \ne \mathrm{An}(\Gamma_2 \circ \Gamma_1)$, in general. For take Γ_1 corresponding to the double occurrence word *aa* and Γ_2 corresponding to *bbcddc*, then $\mathrm{An}(\Gamma_1 \circ \Gamma_2) = 1$ and $\mathrm{An}(\Gamma_2 \circ \Gamma_1) = 2$. See Example 4.5.

Corollary 4.8. $|\mathrm{An}(\Gamma_1 \circ \Gamma_2) - \mathrm{An}(\Gamma_2 \circ \Gamma_1)| \le 1$.

Proof. By Lemma 4.6, each of $\mathrm{An}(\Gamma_1 \circ \Gamma_2)$ and $\mathrm{An}(\Gamma_2 \circ \Gamma_1)$ is equal to $\mathrm{An}(\Gamma_1) + \mathrm{An}(\Gamma_2)$ or $\mathrm{An}(\Gamma_1) + \mathrm{An}(\Gamma_2) - 1$; hence $|\mathrm{An}(\Gamma_1 \circ \Gamma_2) - \mathrm{An}(\Gamma_2 \circ \Gamma_1)| \le 1$. □

Proposition 4.9. *For any positive integer n, there exists*

(i) *a reducible assembly graph Γ such that* $\mathrm{An}(\Gamma) = n$,
(ii) *an irreducible assembly graph Γ such that* $\mathrm{An}(\Gamma) = n$, *and*
(iii) *an assembly graph Γ with no endpoints such that* $\mathrm{An}(\Gamma) = n$.

Proof. (i) Let n be a positive integer. Let Γ_u be the assembly graph with two endpoints that corresponds to the double occurrence word $u = v_1v_2v_2v_1v_3v_3$ (this is the subgraph containing vertices v_1, v_2, v_3 from the graph depicted in Fig. 5(B)). Consider the n-fold composition $\Gamma_n' = \Gamma_u \circ \cdots \circ \Gamma_u$ of Γ_u. Let w_n be the double occurrence word that corresponds to Γ_n', and let Γ_n be the simple assembly graph with two endpoints that corresponds to the word $v_0v_0w_n$ (for $n = 1$, Γ_1 is depicted in Fig. 5(B) and $\mathrm{An}(\Gamma_1) = 2$). Then by induction on n and using a similar argument as in the proof of Lemma 4.6, it follows that $\mathrm{An}(\Gamma_n) = n + 1$.

(ii) To prove the proposition in an irreducible case, observe that it is sufficient to "find" Γ_n described above embedded in an irreducible graph. Consider the double occurrence word *zuz*, where *u* is the double occurrence word that corresponds to Γ_n. Let Γ_{zuz} be the assembly graph that corresponds to *zuz*. Clearly, *zuz* is irreducible, and $\mathrm{An}(\Gamma_{zuz}) = \mathrm{An}(\Gamma_n)$ as the Hamiltonian set for Γ_{zuz} can be obtained from the Hamiltonian set for Γ_n by adding the vertex z to the appropriate path in Hamiltonian set for Γ_n.

(iii) The case of an assembly graph with no endpoints follows directly, by observing that the graph Γ_{n+1} with the two endpoints identified has assembly number n. □

Definition 4.10. For a positive integer n, we define *the minimal realization number for n* to be $R_{\min}(n) = \min\{|\Gamma| : \mathrm{An}(\Gamma) = n\}$, where $|\Gamma|$ is the number of 4-valent vertices in Γ. A graph Γ such that $R_{\min}(n) = |\Gamma|$ is called *a realization of* $R_{\min}(n)$.

Example 4.11. Through case by case inspection, we find that $\mathrm{An}(\Gamma) = 1$ for all assembly graphs Γ with $|\Gamma| \le 3$, and the graph Γ in Fig. 5 (B) has $\mathrm{An}(\Gamma) = 2$ and $|\Gamma| = 4$, so we have $R_{\min}(1) = 1$ and $R_{\min}(2) = 4$.

Definition 4.12. Let $\Gamma_1 = (V_1, E_1)$ be an assembly graph. We define the assembly graph Γ_2 *obtained from* Γ_1 *by deleting a vertex* $v \in V_1$ as follows. Let $\gamma = (v_0, e_1, v_1, e_2, \ldots, e_n, v_n)$ be a transverse Eulerian path in Γ_1. Define $\Gamma_2 = (V_2, E_2)$ by setting $V_2 = V_1 - \{v\}$, and $E_2 = (E_1 - \{e_i, e_k, e_{i+1}, e_{k+1}\}) \cup \{l_i, l_k\}$, where $v_i = v_k = v$ $(i + 1 \ne k)$, l_i is a new edge from v_{i-1} to v_{i+1}, and l_k is a new edge from v_{k-1} to v_{k+1}. The order of the new edges l_i and l_k at vertices v_{i-1}, v_{i+1} and v_{k-1}, v_{k+1} is inherited from the edges e_i, e_{i+1} and e_k, e_{k+1} respectively, such that locally the rigid vertices v_{i-1}, v_{i+1}, v_{k-1}, and v_{k+1} remain the same. In the case when $i + 1 = k$, only one new edge connecting v_{i-1} with $v_{i+2} = v_{k+1}$ is added. The new edge l_i is denoted by $l_i = \widehat{e_iv_ie_{i+1}}$. See Fig. 6.

It is clear that Γ_2 does not depend on the choice of γ.

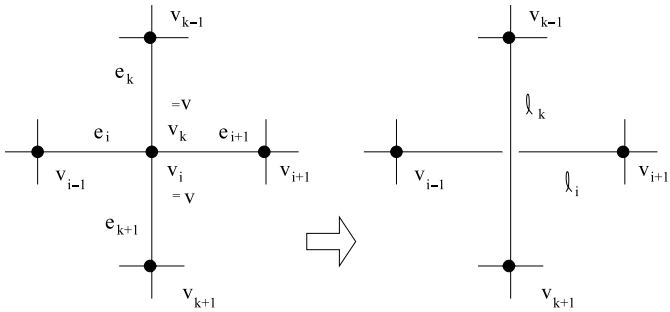

Fig. 6. Removing a vertex $v = v_i = v_k$, the case when $i + 1 \neq k$.

Proposition 4.13. *The following properties hold for* $R_{\min}$.

(i) *For every positive integer n,* $R_{\min}(n) < R_{\min}(n+1)$.
(ii) *If* $R_{\min}(n) = k$ *for some n and k, then for every* $s \geq k$ *there is an assembly graph* Γ *with s 4-valent vertices such that* $\mathrm{An}(\Gamma) = n$.
(iii) $R_{\min}(n) \leq 3(n-1)+1$ *for every positive integer n.*

Proof. (i) Let $R_{\min}(n+1) = k$ for a positive integer k. Let Γ_1 be an assembly graph with k 4-valent vertices such that $\mathrm{An}(\Gamma_1) = n+1$, i.e., Γ_1 is a realization of $R_{\min}(n+1)$. Let Γ_2 be a graph obtained from Γ_1 by deleting a vertex v with or without a loop. Since Γ_2 has one less vertex than Γ_1, $\mathrm{An}(\Gamma_2) < n+1$. Suppose $\mathrm{An}(\Gamma_2) = m < n$ and there is a Hamiltonian set for Γ_2 with m components that does not traverse any of the new edge(s) in Γ_2 added in the process of removal of v. Then by adding to this Hamiltonian set of m components for Γ_2 a short arc containing v and parts of two neighboring edges in Γ_1, a Hamiltonian set of paths with $m+1$ components for Γ_1 is created. But, $m+1 < n+1$, which contradicts the assumption that Γ_1 is a realization of $R_{\min}(n+1)$. Hence $\mathrm{An}(\Gamma_2) = n$.

Next we consider the case when every Hamiltonian set traverses at least one of the new edges in Γ_2 added in the process of removal of v. We show the case when v does not have a loop. The case when v has a loop can be shown in a similar manner. Let $E(v) = \{v_1, v_2, v_3, v_4\}$ in Γ_1, and denote with e_i the edges connecting v with v_i ($i = 1, \ldots, 4$). We observe that at least two of the vertices in $E(v)$ must be distinct. If all are equal, $v_1 = v_2 = v_3 = v_4 = w$, then Γ_1 consists of two 4-valent vertices, and it is not a realization of any $R_{\min}(n)$ by Example 4.11. Further, if three of the vertices are equal, say $v_1 = v_2 = v_3 = w$, then there are three parallel edges connecting v and w, and therefore Γ_1 is not a simple assembly graph. Without loss of generality, we can assume that in Γ_2 there is an edge e connecting v_1, v_3 and another edge f connecting v_2, v_4 (where one or both of these pairs of vertices may be in fact the same vertex). By assumption every Hamiltonian set for Γ_2 contains paths that traverse one or both of these edges entirely. Suppose only one of these edges is traversed, say edge e connecting vertices v_1 and v_3. Then let $\gamma = \gamma_1 v_1 e v_3 \gamma_2$ be the one component in the Hamiltonian set Δ for Γ_2 that traverses e. A Hamiltonian set for Γ_1 is constructed with $|\Delta| + 1 = m + 1$ components with $\Delta' = \Delta \setminus \{\gamma\} \cup \{\delta_1, \delta_2\}$, where $\delta_1 = \gamma_1 v_1 e_1 v e_2'$ and $\delta_2 = e_3' v_3 \gamma_2$ are such that e_2' is a proper segment of e_2 with boundary vertex v, and e_3' is a proper segment of e_3 with boundary vertex v_3. The case when both edges e and f are traversed by every Hamiltonian set of Γ_2 with m components can be shown similarly. Let $\gamma = \gamma_1 v_1 e v_3 \gamma_3$ and $\gamma' = \gamma_2 v_2 f v_4 \gamma_4$ be two components of a Hamiltonian set Δ for Γ_2 traversing both edges e and f. If γ and γ' represent the same component, without loss of generality we can assume that γ_1, γ_2 have the same endpoints and γ_3, γ_4 have the same endpoints, i.e., the component (polygonal path) connects vertices v_1 and v_4 with a portion $\hat{\gamma}$ without visiting vertices v_2 and v_3. Then $\gamma = \gamma_2 v_2 f v_4 \hat{\gamma} v_1 e v_3 \gamma_3 = \gamma'$. We form a Hamiltonian set Δ' for Γ_1 with $|\Delta| + 1 = m + 1$ components with $\Delta' = \Delta \setminus \{\gamma\} \cup \{\delta_1, \delta_2\}$, where $\delta_1 = e_4' v_4 \hat{\gamma} v_1 e_1 v e_2 v_2 \gamma_2$ and $\delta_2 = e_3' v_3 \gamma_3$ are such that e_4' is a proper segment of e_4 with boundary vertex v_4, and e_3' is a proper segment of e_3 with boundary vertex v_3. When γ and γ' are separate components, $\Delta' = \Delta \setminus \{\gamma, \gamma'\} \cup \{\delta_1, \delta_2, \delta_3\}$ with $\delta_1 = e_4' v_4 \gamma_4$, $\delta_2 = \gamma_1 v_1 e_1 v e_2 v_2 \gamma_2$ and $\delta_3 = e_3' v_3 \gamma_3$. Because Γ_1 is a realization of $R_{\min}(n+1)$, it is not possible to have $m < n$; therefore $\mathrm{An}(\Gamma_2) = n$. Hence, $R_{\min}(n) \leq k - 1 < k = R_{\min}(n+1)$.

(ii) Let $R_{\min}(n) = k$ and let Γ_w be an assembly graph that corresponds to a double occurrence string w such that $\mathrm{An}(\Gamma_w) = n$ and $|w| = k$. There is a Hamiltonian set $S = \{\gamma_1, \gamma_2, \ldots, \gamma_n\}$ of polygonal paths for Γ_w. Let q be the last vertex visited by one of the paths in S, say γ_1. For any positive integer s one forms a double occurrence word u from w by $u = w_1 q p_1 p_1 p_2 p_2 \cdots p_s p_s w_2$, where $w = w_1 q w_2$, is the second occurrence of the symbol q in w, and $p_1, p_2, \ldots, p_s$ are symbols that do not appear in w. Then a Hamiltonian set S' for Γ_u can be obtained as $S' = \{\gamma_1', \gamma_2, \ldots, \gamma_n\}$, where γ_1' is obtained by extending γ_1 through vertices $p_1, \ldots, p_s$ in a polygonal manner. Consequently, $\mathrm{An}(\Gamma_u) = n$ and $|u| = k + s$. Hence the statement (ii) is true.

(iii) For the graph Γ_n in the proof of Proposition 4.9, the number of 4-valent vertices is $3(n-1)+1$ and $\mathrm{An}(\Gamma) = n$. Hence $R_{\min}(n) \leq 3(n-1)+1$. □

For two words v, w, we write $w \subset v$ if, as a sequence of symbols, w is a subsequence of v. We say that a word w is *realizable* if Γ_w is realizable. Otherwise it is called *unrealizable*.

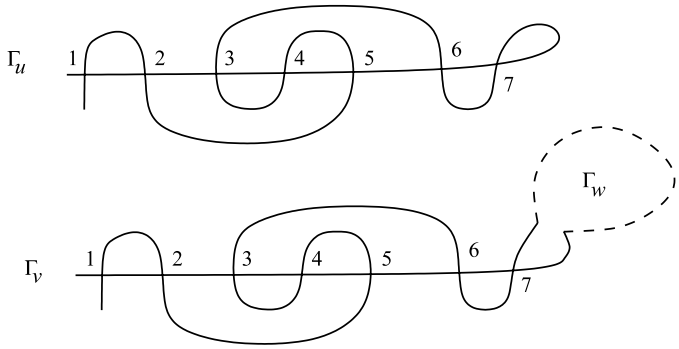

Fig. 7. Unrealizable graphs.

Proposition 4.14. *There exists a constant N such that, for any realizable w, there exists an unrealizable irreducible v such that $w \subset v$ and $|v| - |w| < N$.*

Proof. By brute force examination one can check that the following word is not realizable: $u = 12345677634521$. The corresponding assembly graph Γ_u is depicted in Fig. 7.

Let w be any realizable double occurrence word. Then $v = 1234567w7634521$ (see Fig. 7 for a sketch of Γ_v) is a non-realizable double occurrence word and assume that the symbols in w are distinct from $\{1, \ldots, 7\}$. For if there were a Hamiltonian polygonal path in Γ_v, then this path would start either in Γ_w or out of Γ_w. Note that Γ_w as an assembly graph is a subgraph of Γ_v. A Hamiltonian polygonal path visits every vertex exactly once, so, once a path enters Γ_w, it has to visit all vertices in Γ_w before it exists through vertex 7. If a Hamiltonian path starts in Γ_w, then first it has to visit every vertex in Γ_w and then it exits Γ_w through vertex 7. Since u is not realizable, the rest of the path cannot visit every vertex in $\{1, \ldots, 7\}$ and therefore cannot visit every vertex in Γ_v. If the path starts outside of Γ_w, in order to visit vertices of Γ_w, the path must pass vertex 7. However, since u is not realizable, the path cannot visit every vertex from $\{1, \ldots, 7\}$ before visiting a vertex in Γ_w. Hence, for any realizable w, there exists an unrealizable irreducible v such that $w \subset v$ and $|v| - |w| < 8$.

Corollary 4.15. *There exists a constant N such that, for any w such that $\mathrm{An}(\Gamma_w) = k$ for some positive integer k, there exists an irreducible double occurrence word v with $w \subset v$, such that $\mathrm{An}(v) = k + 1$, and $|v| - |w| < N$.*

Proof. This is essentially the same as the proof of Proposition 4.14 by substituting the realizable word w with a word that defines a graph Γ with $\mathrm{An}(\Gamma) = k$. □

Corollary 4.16. *For any positive integer k, there exists a constant N_k such that, for any realizable w, there exists an unrealizable v such that $\mathrm{An}(v) = k$, $w \subset v$, and $|v| - |w| < N_k$. (One can take $N_k = 3 + 5(k - 1)$.)*

The above Proposition 4.14 and its two corollaries show that if we take the difference between the number of vertices of assembly graphs as a measure of "distance" between the graphs, then the results can be interpreted as follows. (i) For every realizable assembly graph Γ, there is an unrealizable word within a radius of 8 from Γ. (ii) For every assembly graph Γ, there is an unrealizable graph within radius 8 from Γ. (iii) For every realizable graph Γ, there is an assembly graph with assembly number k within radius $3 + 5(k - 1)$ from Γ.

Proposition 4.17. *For any unrealizable double occurrence word v with $|v| = m$, there exist a constant $N(m)$ and a realizable double occurrence word w such that $v \subset w$, and $|w| - |v| \leq N(m)$.*

Proof. Let v be an unrealizable double occurrence word, and let Γ_v be the assembly graph that corresponds to v. If $\mathrm{An}(\Gamma_v) = k$, then there is a Hamiltonian set of polygonal paths $S = \{\gamma_1, \gamma_2, \ldots, \gamma_k\}$ for Γ_v. Let v_{i1} and v_{i2} be the initial and terminal vertices visited by the path γ_i for $i \in \{1, 2, \ldots, k\}$. Construct w from v as follows: add a symbol p_i after the first appearance of v_{i1} and before the first appearance of $v_{(i+1)2}$ in v for $i \in \{1, 2, \ldots, k - 1\}$ such that $p_i \neq p_j$ when $i \neq j$ and the p_i's are different from any symbol in v. By construction we have that $v \subset w$. Note that, in Γ_w, for each i, the path $v_{i2}p_iv_{(i+1)2}$ is polygonal, and therefore, the path $\gamma_1, v_{12}, p_1, v_{21}, \gamma_2, v_{22}, p_2, v_{31}, \gamma_3, \ldots, v_{2(k-1)}, p_{k-1}, v_{1k}, \gamma_k$ is a Hamiltonian polygonal path in Γ_w. So w is realizable. Note that there are $m = k - 1$ new symbols added in w; hence we can take $N(m) = k + 1$. □

Remark 4.18. The preceding proposition holds with a requirement that v is also irreducible. In other words, for any unrealizable irreducible double occurrence word v with $|v| = m$, there exist a constant $N(m)$ and a realizable double occurrence word w such that $v \subset w$, and $|w| - |v| \leq N(m)$. To see this let v' be the double occurrence word obtained from v by adding two more symbols q_1 and q_2: $v' = q_1w_1q_2w_2q_2q_1$, where $w_1w_2 = v$ and the first occurrence of q_2 is positioned right before the first occurrence of v_{11}. Then v' is our desired word. In this case, the constant $N(m)$ increases by 1.

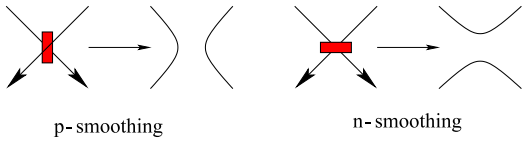

Fig. 8. Two types of smoothing: parallel (p-)smoothing (left) and non-parallel (n-)smoothing (right).

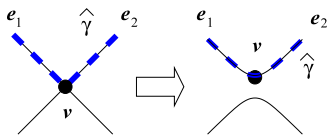

Fig. 9. Smoothing at a vertex v with respect to a polygonal path $\hat{\gamma}$.

5. Smoothings of assembly graphs

As mentioned earlier, it is assumed that the DNA recombination appears when pointer sequences are aligned. Alignment of the pointers is represented with a 4-valent vertex in an assembly graph. The recombination, as a result of splicing, is represented as a smoothing of the vertex (see below). If the pointer sequences are aligned in parallel, then the smoothing of the vertex follows the predetermined direction of the graph, called below parallel smoothing. If the pointer alignment is antiparallel, then the smoothing of the vertex is performed incompatibly with the direction of the graph, called below non-parallel smoothing. In this section we define smoothing of assembly graphs determined by a polygonal path and compare this notion with the existent notion of smoothing in virtual knot diagrams with proper coloring as defined in [7].

Let Γ be a directed simple assembly graph with rigid 4-valent vertices $V = \{v_1, \ldots, v_m\}$ corresponding to a double occurrence word w. Assume that the word w is consistent with the given direction, and in particular, we have $\mathrm{dom}(w) = V$. A *smoothing* of a 4-valent vertex is defined in a similar manner as a smoothing of a crossing of a knot diagram (see, for example, [6]); a neighborhood of a vertex is removed and two parallel arcs are attached as depicted in Fig. 8. Each smoothing at a crossing is either orientation preserving (*parallel smoothing*, or *p-smoothing*, as in Fig. 8 left) or non-preserving (*non-parallel smoothing*, or *n-smoothing* as in Fig. 8 right). Fig. 8 also depicts a convention of placing thin rectangles, called markers, at 4-valent vertices to indicate two types of smoothing. Although we start with a directed simple assembly graph, if n-smoothings are performed at some of the vertices, there is no direction defined for the resulting graph.

One can see smoothing at a vertex v as a splitting of the vertex v to two 2-valent vertices, such that the pairs of the edges incident to the same vertex after the smoothing are neighbors at v. An open neighborhood of a 2-valent vertex is homeomorphic to an open interval; hence, such vertices are depicted by arcs, as shown in Fig. 8.

The sets of Hamiltonian paths can be related to sets of smoothings as follows. Let Γ be an assembly graph. Let $\gamma = \{\gamma_1, \ldots, \gamma_k\}$ be a Hamiltonian set of polygonal paths for Γ.

Definition 5.1. A smoothing of a 4-valent vertex v in Γ with respect to a polygonal path $\gamma_i \in \gamma$ is a smoothing such that the neighboring edges traversed by γ_i at v remain connected after the smoothing (see Fig. 9).

A *smoothing of Γ with respect to a Hamiltonian set of polygonal paths* $\gamma = \{\gamma_1, \ldots, \gamma_k\}$ is the graph denoted $\tilde{\Gamma}_\gamma$ obtained after smoothing of every 4-valent vertex v with respect to the path from γ that visits v.

Note that the smoothing $\tilde{\Gamma}_\gamma$ consists of arcs and closed curves (topologically, circles).

Remark 5.2. Let Γ be an assembly graph, $\gamma = \{\gamma_1, \ldots, \gamma_k\}$ be a Hamiltonian set of polygonal paths, and $\tilde{\Gamma}_\gamma$ be the smoothing of Γ with respect to γ. Since, for each $i = 1, \ldots, k$, neighboring edges of γ_i remain connected after smoothing, a path corresponding to γ_i is naturally defined in $\tilde{\Gamma}_\gamma$ as depicted by a thick dotted line in the upper arc in Fig. 9 to the right. We denote these paths using the same notation $\{\gamma_1, \ldots, \gamma_k\}$ whenever it is clear from the context. Hence, after smoothing, the new set of paths $\gamma = \{\gamma_1, \ldots, \gamma_k\}$ is a subset of $\tilde{\Gamma}_\gamma$. Furthermore, on each arc or circle in $\tilde{\Gamma}_\gamma$, the edges and vertices (their "traces," or "scars") can be regarded as kept intact, as depicted in Fig. 9 in the right. We also continue to use the same symbols for these vertices and edges when regarded as subsets of arcs and circles in $\tilde{\Gamma}_\gamma$ and call them *vertices and edges inherited from Γ*.

As noted in Section 1, an assembly graph can be seen as a physical representation of the DNA with the recombination sites represented by crossings (4-valent vertices). The recombination is obtained by splicing the DNA at the recombination sites. In order for the resulting structure to be biologically functional, the resulting structure has to be unlinked (unknotted). In the following we show that every assembly graph Γ, given with a Hamiltonian set of polygonal paths γ, can be embedded in $\mathbb{R}^3$ such that $\tilde{\Gamma}_\gamma$ is unlinked. This implies that, for every set of rearrangements of a given DNA sequence, the molecule can be positioned in space such that, after all rearrangements are performed (some or all simultaneously), the resulting sequence encoding a functional gene can "easily" be identified and separated from the other pieces.

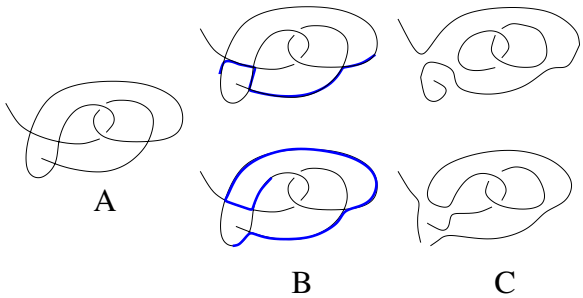

Fig. 10. Two Hamiltonian polygonal paths giving distinct smoothed embeddings.

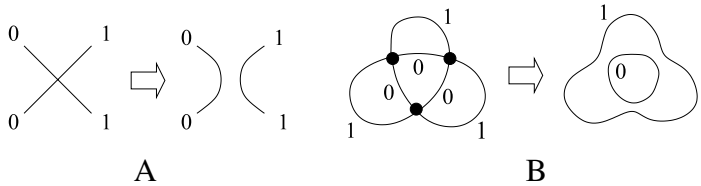

Fig. 11. (A) Proper coloring of a vertex and smoothing corresponding to the proper coloring (Kauffman [7]). (B) An example of a proper coloring for an assembly graph "representing" the trefoil and the resulting components after the smoothing.

Proposition 5.3. *For any assembly graph Γ and a Hamiltonian set of polygonal paths $\gamma = \{\gamma_1, \ldots, \gamma_k\}$, there is an embedding of Γ in $\mathbb{R}^3$ such that $\tilde{\Gamma}_\gamma$ is unlinked.*

Proof. Consider a generic map from Γ to the plane $\mathbb{R}^2$. Aside from the 4-valent vertices, such a map immerses edges from Γ with possible transverse double points. We need to determine the crossing information for each such transverse intersection, so that the result can be regarded as a diagram of an embedding of Γ in $\mathbb{R}^3$. To the image of Γ in $\mathbb{R}^2$ corresponds an image of $\tilde{\Gamma}_\gamma$ in $\mathbb{R}^2$ such that the 4-valent vertices of Γ are smoothed and the transverse double points from the image of Γ coincide with the transverse double points of $\tilde{\Gamma}_\gamma$. Denote the (circular) components of $\tilde{\Gamma}_\gamma$ by $c_1, c_2, \ldots, c_\ell$. Since smoothings are performed locally at each 4-valent vertex, $\{c_1, c_2, \ldots, c_\ell\}$ can be regarded as immersed, (closed) plane curves. Choose base points $b_1, \ldots, b_\ell$ on $c_1, \ldots, c_\ell$, respectively. Each component c_i is traced in one direction, starting from point b_i. The crossing information at transverse double points is given in an descending order; the arc visited first is the over-arc, and the arc visited the second time is the under-arc. If a crossing is between two components c_i and c_j for $i < j$, then the arc being part of c_i is designated as an over-arc. It is well known in knot theory that such a descending order gives a diagram of the unlink [10]. Now we take the same crossing information at the transverse double points of the image of Γ to Γ itself, and we obtain a desired embedding. □

Example 5.4. There exists an embedding of a simple assembly graph Γ in $\mathbb{R}^3$, and two Hamiltonian polygonal paths γ_1 and $\gamma_2 \subset \Gamma$, such that for the corresponding smoothings $\tilde{\Gamma}_{\gamma_1}$ and $\tilde{\Gamma}_{\gamma_2}$ are distinct; see Fig. 10. Note also that this provides an example of an assembly graph with two inequivalent Hamiltonian polygonal paths.

The above example raises a natural question: Can an assembly graph Γ be embedded in $\mathbb{R}^3$ such that, for any Hamiltonian set of paths, $\tilde{\Gamma}_\gamma$ is unlinked?

Smoothings of knot diagrams corresponding to proper colorings have been studied by Kauffman [7]. A *proper coloring* is an assignment of two colors 0 and 1 to the edges of an assembly graph such that non-neighboring edges are colored distinctly. The smoothing with respect to a proper coloring is performed such that the neighboring edges colored with the same color remain connected (see the left of Fig. 11 (A)). For a given proper coloring of an assembly graph, there is a unique smoothing at every crossing such that the resulting circles have consistent colors, 0 or 1, as depicted in the right of Fig. 11 (A). In Fig. 11 (B), such a smoothing is depicted for a projection of the trefoil knot. An invariant for virtual knots through can be defined through "counting" of proper colorings [7].

Here we observe that smoothings through Hamiltonian sets of polygonal paths define different operation than the one defined by smoothings by proper colorings, such that the number of resulting components can be different. A proper coloring for assembly graphs are defined in the same manner as for knot diagrams, regarding 4-valent vertices as crossings of knot diagrams, and imposing the same coloring condition at every vertex. It is known (see [7]) that for a planar simple assembly graph with no endpoints (having one transverse Eulerian path) and with a given a direction, there is a unique smoothing for a proper coloring; the smoothing obtained when at every crossing the parallel smoothing is performed (Fig. 8 left) with respect to the orientation. Consequently, simple assembly graphs with no endpoints have exactly two proper colorings: one choice of 0 and 1 and the other (reversed) choice. For example, the trefoil projection in Fig. 11 (B) has only one other proper

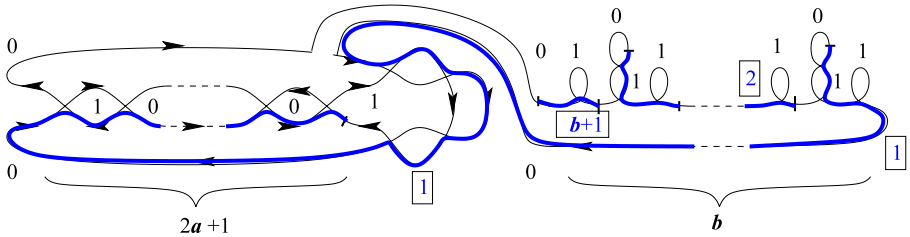

Fig. 12. An example showing the relationship between Kauffman's smoothings and smoothings by polygonal paths. The depicted graph Γ has $\mathrm{An}(\Gamma) = m = b + 1$ and $K_{\min} = n = a + b + 1$.

coloring: the inner circle colored by 1 and the outer circle colored by 0. The situation for virtual knots and links is more complex, and is characterized completely in [7].

Definition 5.5. Let $\mathcal{C}$ be the set of proper colorings for an assembly graph Γ with one or more transverse components and no endpoints. Given a coloring $\xi \in \mathcal{C}$, let ξ_0 and ξ_1 denote the number of components colored 0, respectively 1, after smoothing corresponding to ξ. The minimal number of components of smoothed circles with one color is defined to be $K_{\min} = \min_{\xi\in\mathcal{C}}\{\xi_0, \xi_1\}$.

Each smoothing with a proper coloring defines a set of Hamiltonian polygonal paths; take the set of circles colored by 0, say, and remove a point from each circle component to obtain open paths. The color with smaller number of components give rise to the smaller number of Hamiltonian polygonal paths; hence we have the following.

Lemma 5.6. *For any assembly graph Γ with no endpoints,* $\mathrm{An}(\Gamma) \le K_{\min}(\Gamma)$.

Other than this inequality, however, these numbers are independent in the following sense.

Proposition 5.7. *For any positive integer m and n with $m \le n$, there is an assembly graph Γ with no endpoints such that* $\mathrm{An}(\Gamma) = m$ *and* $K_{\min}(\Gamma) = n$.

Proof. Let Γ be the assembly graph depicted in Fig. 12. There are $2a + 1$ of 4-valent vertices to the left, attached to Γ_b to the right of the graph, which consists of b copies of the assembly graph in Fig. 5(B) (see proof of Proposition 4.9). The numbers a and b are non-negative integers that are to be determined for given m and n. From the proof of Lemma 4.6, we have $\mathrm{An}(\Gamma) = b + 1$, and a minimal set of Hamiltonian polygonal paths is depicted in Fig. 12 by thick lines. The components of the Hamiltonian set are labeled by boxed numbers, $\boxed{1}, \ldots, \boxed{\text{b+1}}$ A direction of Γ is indicated by arrows, and a choice of proper coloring with 0 and 1 is labeled in the figure. A smoothing with respect to the corresponding proper coloring is determined by p-smoothings at all vertices. There are $a + b + 1$ circles colored by 0 and $a + 3b + 1$ circles colored by 1. Since this is a planar diagram, this is the only result of smoothing with proper colorings [10]. Therefore $K_{\min}(\Gamma) = a + b + 1$. Thus we obtain the result by the choice determined by $b + 1 = m$ and $a + b + 1 = n$; that is, $a = n - m$ and $b = m - 1$. □

6. Successful simultaneous smoothings

As mentioned earlier, each polygonal path in an assembly graph can be seen as a unique representation of a macronuclear gene. A few rearrangements may occur simultaneously or within a short time frame difference to be considered as simultaneous. We assume that the recombination that occurs simultaneously does not disturb the natural MDS order in the macronuclear gene. In our model of assembly graph, a set of simultaneous rearrangements translates into a set of simultaneous smoothings, and the consistent order of the MDSs during the recombination process becomes a requirement that the corresponding polygonal path remains connected after these smoothings. In this section we characterize the sets of vertices in a simple assembly graph that allow simultaneous smoothings and provide a way to obtain smoothing strategies (i.e., recombination strategies) for a given simple assembly graph.

Let Γ be a simple assembly graph, and let $\tilde{\Gamma}_\gamma$ be the smoothed graph of Γ with respect to a Hamiltonian polygonal path γ. Note that $\tilde{\Gamma}_\gamma$ does not contain any 4-valent vertices, and might have two vertices of degree 1 equivalent to the endpoints of Γ. The graph $\tilde{\Gamma}_\gamma$ can have more than one component. If Γ has endpoints then one of the components in $\tilde{\Gamma}_\gamma$ is an arc component and the rest are closed curves (topologically circles).

Definition 6.1. Let Γ be a realizable assembly graph (i.e., $\mathrm{An}(\Gamma) = 1$), and S a subset of vertices in Γ. An *S-partial smoothing of Γ with respect to a Hamiltonian polygonal path γ* is an assembly graph with a set of 4-valent vertices $V(\Gamma) \setminus S$, denoted by $\tilde{\Gamma}_{(\gamma,S)}$, obtained by smoothing of all vertices in S with respect to γ.

Definition 6.2. Let Γ be a realizable assembly graph and γ a polygonal Hamiltonian path. A subset $S \subset V(\Gamma)$ is called *linear* with respect to γ if the S-partial smoothing of Γ with respect to γ is a simple assembly graph.

Note that $\emptyset \subset V(\Gamma)$ is linear with respect to any polygonal path γ iff Γ is a simple assembly graph. Let $S \subset V(\Gamma)$. Recall the process of deleting a vertex in Γ (Definition 4.12). We denote with Γ_S the assembly graph obtained from Γ by deleting all of the vertices from $V(\Gamma) \setminus S$. Note that the process of deletion of a vertex is local, i.e., Γ is changed at a local neighborhood of the deleted vertex without affecting the structural rigidity of the other vertices. Hence if a set T of vertices is deleted from Γ, then this can be done by fixing an order of the vertices in T, and then removing vertices from T one by one according to the order. Since the removal of one vertex does not affect the local structure of the others, the resulting graph is invariant with respect to the choice of the order of vertices in T.

Remark 6.3. If Γ is a simple assembly graph, with or without endpoints, removal of all vertices in Γ results in a single component diagram (arc if Γ has endpoints, or circle if Γ has no endpoints). This follows because the transverse Eulerian path in Γ determines the single component.

Let γ and γ' be two polygonal paths in the simple assembly graph Γ. Let both γ and γ' contain the 4-valent vertex v. Denote by e_1, e_2, e_3, e_4 the incident edges of v. Let e_1, v, e_2 be subsequence of γ. Then, we say that γ' *coincides with* γ at vertex v if e_1, v, e_2 or e_2, v, e_1 is a subsequence of γ', and we say that γ' is *disjoint to* γ at vertex v if e_3, v, e_4 or e_4, v, e_3 is a subsequence of γ'.

We can characterize linear subsets as follows.

Proposition 6.4. *A subset $S \subset V(\Gamma)$ is linear with respect to a polygonal Hamiltonian path γ iff Γ_S has a polygonal path γ' that is Eulerian (i.e., visiting every edge exactly once), and at every vertex in S the path γ' either coincides with γ or is disjoint to γ (i.e., it traces the other two edges incident to the vertex).*

Proof. Let Γ be a simple assembly graph with or without endpoints having a Hamiltonian polygonal path γ. Assume that the set of vertices S is linear. Denote with S^c the set $V(\Gamma) \setminus S$. By performing smoothings at vertices in S with respect to γ, we obtain the simple assembly graph $\tilde{\Gamma}_{(\gamma,S)}$ with the set of vertices S^c. Now the removal of the vertices from S^c means removal of all of the vertices from the simple assembly graph $\tilde{\Gamma}_{(\gamma,S)}$. The removal of each vertex is done such that the new edges are denoted with the pair of edges that is removed, i.e., if a vertex v_i is removed and the incident edges e_i and e_{i+1} are substituted with a new edge l_i, we write $l_i = (e_i, v_i, e_{i+1})$ (see Fig. 6). The resulting graph Γ', by Remark 6.3, is a single component diagram (arc or a circle). This component is labeled by the sequence of edges and vertices determining the required Eulerian path in Γ_S. It is obtained by tracing the single component of Γ' (either from one endpoint to the other, or from a chosen vertex to a return to that vertex). In this path every pair of edges incident to a vertex in S must either coincide with γ or is disjoint from γ since Γ' was obtained after smoothing all of the vertices in S with respect to γ.

Conversely, let γ' be a Eulerian polygonal path in Γ_S which at every vertex in S either coincides with γ or is disjoint to γ.

Such a path can be considered as a path in Γ, simply by replacing each edge l_i in γ' by e_i, v_i, e_{i+1}, where l_i is obtained by connecting e_{i+1} and e_i after removal of v_i in Γ_S (see Fig. 6). Call this path γ''. As a path in Γ, γ'' is Eulerian since each edge of Γ appears exactly once as part of an edge in Γ_S, and γ' is Eulerian. At a vertex in S, γ'' either coincides with γ, or follows the other two edges, and at a vertex not in S, it coincides with the transverse path. Next consider $\tilde{\Gamma}_{(\gamma,S)}$. If $v \in S$, then v is smoothed in $\tilde{\Gamma}_{(\gamma,S)}$ as depicted in Fig. 9. Hence γ'' can be considered as a path in $\tilde{\Gamma}_{(\gamma,S)}$. This can be done if γ'' either coincides with γ (as depicted in Fig. 9 to the right by a dotted arc) or if γ'' follows the other two edges complementing γ (in which case γ'' is taking the arc opposite of the dotted arc in Fig. 9). Let γ''' be the path in $\tilde{\Gamma}_{(\gamma,S)}$ thus obtained. Since γ'' coincides with the transverse path in Γ at vertices not in S, γ''' is a transverse path in $\tilde{\Gamma}_{(\gamma,S)}$. Furthermore, γ''' is Eulerian, since γ'' is Eulerian. Hence γ''' is a transverse Eulerian path in $\tilde{\Gamma}_{(\gamma,S)}$, which implies that $\tilde{\Gamma}_{(\gamma,S)}$ is a simple assembly graph. Therefore, S is linear. □

Definition 6.5. Let Γ be a realizable assembly graph, and γ be a Hamiltonian polygonal path. A subset $S \subset V(\Gamma)$ is called *successful* with respect to γ if the S-partial smoothing of Γ with respect to γ is an assembly graph that has a transverse arc component containing γ. (Here, we consider that γ remains intact after the smoothing; see Remark 5.2.)

Recall that an assembly graph models the micronuclear DNA such that each 4-valent vertex corresponds to the location where pointers are 'aligned' and the homologous recombination appears. Each smoothing of a 4-valent vertex models the result of the recombination. A (Hamiltonian) polygonal path encodes the result of the recombination such that the MDSs are rearranged in the correct macronuclear linear order. One can think of the edges of the assembly graph as being MDSs or IESs (such a labeling was used in [1]). Thus a successful S-smoothing for a set $S \subset V(\Gamma)$ can be seen as a simultaneous recombination of a set of pointers determined by S. A successful S-smoothing models recombination which keep the MDSs (i.e., the edges from the polygonal path γ) that are expected to be part of an assembled gene on a single molecule. Hence, we regard smoothings of a successful subset S of $V(\Gamma)$ as one possible step in the process of assembly of a functional MAC gene, where recombination determined by the pointers in S occurs simultaneously. It is proven in [1] that, for any simple assembly graph Γ, the set of all vertices, $S = V(\Gamma)$, is successful.

Lemma 6.6. *Let Γ be a simple, realizable assembly graph and γ be a Hamiltonian polygonal path. Then every linear subset S of $V(\Gamma)$ is successful with respect to γ.*

Proof. Let S be a linear subset of $V(\Gamma)$. Then $\tilde{\Gamma}_{(\gamma,S)}$ is a simple assembly graph. Hence $\tilde{\Gamma}_{(\gamma,S)}$ itself is a transverse arc component of $\tilde{\Gamma}_{(\gamma,S)}$ that contains γ. Therefore, S is successful with respect to γ. □

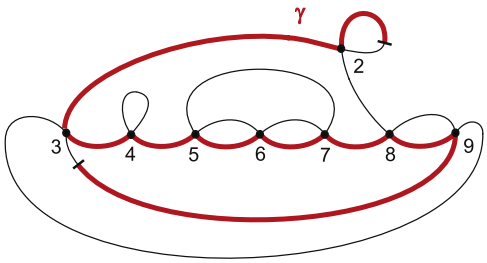

Fig. 13. An assembly graph modeling actin I gene of *S. nova* [1]. A Hamiltonian path in this assembly graph is indicated with a thickened line.

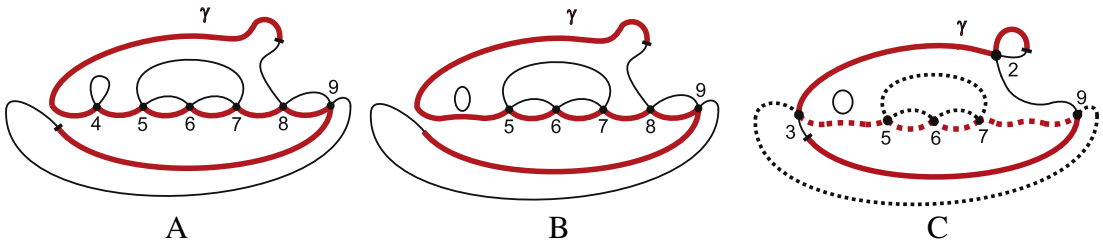

Fig. 14. Three choices of partial smoothings. (A) Example of a linear set: {2, 3}-smoothing of graph Γ from Fig. 13 results in a single transverse component containing γ. (B) Example of a successful but non-linear set: {2, 3, 4}-smoothing of graph Γ from Fig. 13 results in two transverse components, one of which contains γ. (C) Example of a non-linear non-successful set: {4, 8}-smoothing of graph Γ from Fig. 13 results in three transverse components, two of which contain γ. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Example 6.7. Let Γ be the assembly graph depicted in Fig. 13, where a polygonal Hamiltonian path γ is also indicated by a thick line. Consider three sets of vertices $S' = \{2, 3\}$, $S'' = \{2, 3, 4\}$ and $S''' = \{4, 8\}$.

The assembly graphs $\tilde{\Gamma}_{(\gamma,S')}$, $\tilde{\Gamma}_{(\gamma,S'')}$ and $\tilde{\Gamma}_{(\gamma,S''')}$ depicted in Fig. 14(A), Fig. 14(B) and Fig. 14(C), respectively, are obtained from Γ by applying S'-, S''- and S'''-partial smoothings with respect to γ. We observe that S' is linear, since $\tilde{\Gamma}_{(\gamma,S_1)}$ is a simple assembly graph with a Hamiltonian polygonal path indicated with the thick red line. Therefore S' is also successful.

On the other hand, S'' is successful, since $\tilde{\Gamma}_{(\gamma,S'')}$ has a transverse path (i.e., component) that contains γ. However, S'' is not linear, because $\tilde{\Gamma}_{(\gamma,S'')}$ has two transverse components: one containing γ and the other being a separate cyclic component. The set S''' is neither linear nor successful, since $\tilde{\Gamma}_{(\gamma,S''')}$ has three transverse components: one indicated with a dotted line, another indicated with a solid line containing parts of γ, and the third being a small cyclic part which does not contain any part of γ. Since there are two components that contain edges from γ, S''' is neither linear nor successful.

Remark 6.8. The union, intersection, complement, subset or superset of successful sets are not necessarily successful sets. For example, let Γ be the assembly graph and γ a polygonal Hamiltonian path in Γ, as depicted in Fig. 13. Then, the set $\{5, 6, 7\}$ is successful, but $\{5\}$ and $\{5, 6, 7, 8\}$ are not. Also $\{6, 7\}$ and $\{5, 7\}$ are successful, but $\{5, 6, 7\} \setminus \{6, 7\}$ and $\{6, 7\} \cap \{5, 7\}$ are not.

Definition 6.9. Let Γ be a simple realizable assembly graph, and let γ be a Hamiltonian polygonal path $(e_1 \setminus v_0)$, $v_1, e_2, \ldots, v_{m-1}, e_m, v_m, (e_{m+1} \setminus v_{m+1})$ in Γ. A Hamiltonian set of polygonal paths $\{\gamma_1, \gamma_2, \ldots, \gamma_k\}$ is called *complementary* to γ if no edges from γ except (possibly) $(e_1 \setminus v_1)$ and $(e_{m+1} \setminus v_m)$ appear in the paths $\gamma_1, \gamma_2, \ldots, \gamma_k$.

A Hamiltonian set of polygonal paths complementary to γ is called *minimal* if its cardinality is minimal among all Hamiltonian sets of polygonal paths complementary to γ.

Example 6.10. A minimal complementary Hamiltonian set of polygonal paths $\gamma^c = \{\gamma_1, \gamma_2, \gamma_3\}$ for the simple assembly graph Γ and the polygonal Hamiltonian path γ are depicted in Fig. 15.

For the rest of the section, only simple assembly graphs with no endpoints are considered. There is no loss of generality with such an approach, since each simple assembly graph with two endpoints can be transformed into a simple assembly graph with no endpoints by connecting the endpoints. In Lemma 6.11, some properties of complementary Hamiltonian sets of polygonal paths are examined.

Lemma 6.11. *Let Γ be a simple assembly graph with no endpoints,*

$$\gamma = ((e_1 \setminus v_0), v_1, e_2, \ldots, v_{m-1}, e_m, v_m, (e_{m+1} \setminus v_{m+1}))$$

be a Hamiltonian polygonal path in Γ, and $\gamma^c = \{\gamma_1, \gamma_2, \ldots, \gamma_k\}$ be a minimal Hamiltonian set of polygonal paths complementary to γ. The following holds.

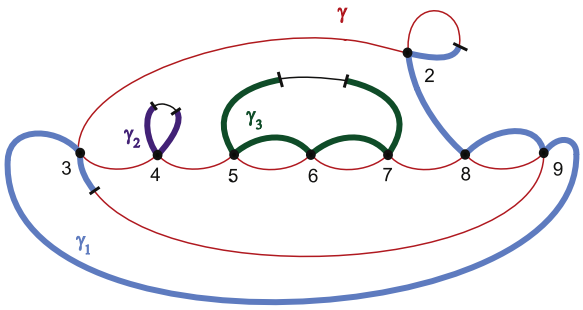

Fig. 15. Complementary paths for the Hamiltonian path depicted in Fig. 13.

(i) *The elements of γ^c are pairwise edge disjoint, i.e., no edge of Γ appears in more than one path from γ^c.*
(ii) *For each edge $e \in E(\Gamma)$, e appears either in γ or there is some i such that e appears in $\gamma_i \in \gamma^c$.*
(iii) *Each $\gamma_i \in \gamma^c$ that does not contain edges from γ starts and ends at the same edge.*
(iv) *If $\gamma_i \in \gamma^c$ contains an edge from γ, then it contains both e_1 and e_{m+1}.*
(v) *γ^c is unique.*

Proof. (i) Let γ_i and γ_j be two distinct polygonal paths in γ^c, each containing a part of an edge e. Let v_1 and v_2 be the endpoints of e (possibly $v_1 = v_2$). Since γ^c is a Hamiltonian set, neither v_1 nor v_2 appears in both γ_i and γ_j. Hence there are three possibilities: (1) each of γ_i, γ_j contains exactly one of v_1 or v_2, (2) both v_1 and v_2 appear in one of the paths γ_i or γ_j, (3) one of v_1 or v_2 does not appear in $\{\gamma_i, \gamma_j\}$. We show that none of these cases holds. If case (1) happens, then assuming v_1 appears in γ_i, $(e \setminus v_2)$ is an initial or terminal edge of γ_i and $(e \setminus v_1)$ is an initial or terminal edge of γ_j. Therefore, the path $\gamma_{ij} = (\gamma_i, e, \gamma_j)$ is a polygonal path in Γ, and $(\gamma^c \setminus \{\gamma_i, \gamma_j\}) \cup \{\gamma_{ij}\}$ is a Hamiltonian set of polygonal paths complementary to γ. This contradicts the minimality of γ^c. Hence, case (1) is not possible. (2) Consider the case when both v_1 and v_2 appear in γ_i. That implies that neither of v_1 and v_2 appears in γ_j, and both $(e \setminus v_1)$ and $(e \setminus v_2)$ are end edges of γ_j and $e \setminus \{v_1, v_2\}$ is an edge in γ_j. Since γ_j is connected, it must consist only of a part of e without the endpoints. But then γ_j is not a polygonal path and it can be removed from γ^c, which violates the minimality condition of γ^c. For case (3), assume, without loss of generality, that v_1 does not appear in $\{\gamma_i, \gamma_j\}$. Then $(e \setminus v_1)$ is an initial or terminal edge of both γ_i and γ_j. If v_2 appears in γ_i, then $(e \setminus v_2)$ is an initial or terminal edge of γ_j, and therefore $\gamma_j = ((e \setminus v_1), (e \setminus v_2))$. Otherwise v_2 is in γ_j but not γ_i, and hence $\gamma_i = ((e \setminus v_1), (e \setminus v_2))$. In either case the resulting path is not a polygonal path, and therefore case (3) is not possible either. Hence γ_i and γ_j are edge disjoint.

(ii) Let $e \in E(\Gamma)$. Note that both γ and γ^c are Hamiltonian. So every vertex v is in both γ and γ_i for some $\gamma_i \in \gamma^c$. Therefore each γ and γ_i contains a pair of neighboring edges incident to v. Hence, if e is a loop at a vertex v, then two pairs of neighboring edges at v must contain e. Suppose e is not a loop and e does not appear in γ. Let v and v' be incident vertices of e. If v (or v') is not v_0 or v_{m+1}, then the four edges incident to v (or v') must belong to two complementary pairs of neighboring edges (one pair belonging to γ and the other to γ_i for some $\gamma_i \in \gamma^c$). Hence e must be in γ_i. The same holds if $e_1 = e_{m+1}$ is incident to v. An edge incident to v belongs to both γ and γ^c only if it is e_1 or e_{m+1} when $e_1 \neq e_{m+1}$. If $v = v_0$ (or $v = v_{m+1}$), the Hamiltonian path γ contains two edges incident to v that are distinct from e_1 (i.e., e_{m+1}), hence one of the paths in γ^c must visit v through e_1 and e (i.e., e_{m+1} and e).

(iii) Let $\gamma_i \in \gamma^c$ be a polygonal path that does not contain edges from γ. Suppose that γ_i starts with the edge e and ends with the edge e' and assume that $e \neq e'$. Let v and v' be the endpoints of e such that γ_i starts with $(e \setminus v)$. Note that if $v = v'$ then e is a loop, i.e., e is a neighbor of itself and γ_i must start and end at e. If v appears in γ_i, then $(e \setminus v')$, v is subsequence of γ_i, implying that γ_i starts and ends at e, so γ_i is circular. This cannot happen if $e \neq e'$. Hence, v does not appear in γ_i. Then, there is some $j \neq i$ such that v appears in γ_j. Therefore $((e \setminus v'), v, e'')$ is a subsequence of γ_j, where e'' is a neighbor of e with respect to v not in γ. But that contradicts (i), since in that case e would be in both γ_i and γ_j. Hence, $e = e'$.

(iv) Note that if $e_1 = e_{m+1}$ then $\gamma_i \in \gamma^c$ cannot contain any edge from γ by definition of γ^c, so, if $\gamma_i \in \gamma^c$ is a polygonal path that contains an edge from γ, $e_1 \neq e_{m+1}$. By the definition of γ^c, γ_i contains e_1 or e_{m+1}. Suppose that γ_i starts with $(e_1 \setminus v_1)$, i.e., $\gamma_i = ((e_1 \setminus v_1), v'_2, \ldots, v'_k(e \setminus v))$. First note that $e_1 \neq e$, since otherwise $v = v_0$ and γ_i contains $(e_1 \setminus v_0)$, which contradicts the definition of γ^c. If $v'_k = v$ then e is a loop. The Hamiltonian path γ visits v either by the two edges incident to v that are distinct from e, or it contains e and one edge distinct from e. In the former case $\gamma_i = ((e \setminus v), v, (e \setminus v))$ and cannot start with $(e_1 \setminus v_1)$. So it must be that γ contains e, in which case it contains $(e \setminus v)$, i.e., $(e \setminus v) = (e_{m+1} \setminus v_{m+1})$.

If $v'_k \neq v$, consider the other three incident edges of v, say e', f, f'. Assume e is a neighbor to both e' and f with respect to v. Since v is not visited by γ_i, there must be $\gamma_j \in \gamma^c$ that contains v. By (i), γ_j cannot contain e, therefore it contains either f, f' or e', f. In both cases the Hamiltonian γ has to visit v containing edges e, e' or e, f. Since e is in both γ and γ_i, by the definition of γ^c, it must be that it is the end of γ, i.e., $v = v_m$ and $v'_k = v_{m+1}$.

(v) First, note that there is no element of γ^c that contains edges e_1 or e_{m+1} from γ if $e_1 = e_{m+1}$. If $e_1 \neq e_{m+1}$, then by (iv), there is exactly one element of γ^c that contains both e_1 and e_{m+1}. Let $\gamma^c = \{\gamma_1, \ldots, \gamma_k\}$ and $\hat{\gamma}^c = \{\gamma'_1, \ldots, \gamma'_{k'}\}$ be two distinct minimal Hamiltonian sets of polygonal paths complementary to γ. From the above argument, both γ^c and $\hat{\gamma}^c$ contain either one or no element that has edges from γ. Now, let us show that the paths that do not contain edges from

γ are the same in both γ^c and $\hat{\gamma}^c$. Let $\gamma_i = ((e_{1i} \setminus v_{0i}), v_{1i}, e_{2i}, \ldots, v_{s-1i}, e_{si}, v_{si}, (e_{s+1i} \setminus v_{s+1i}))$ be an element of γ^c that does not contain edges from γ. By (iii) we have that $e_{1i} = e_{s+1i}$, i.e., γ_i is a cycle and $v_{0i} = v_{si}$, $v_{1i} = v_{s+1i}$. There is some $j \in \{1, \ldots, k'\}$ such that $\gamma_j' \in \hat{\gamma}^c$ is a polygonal path that visits the vertex v_{1i}. Then e_{2i} appears in γ_j'. If γ_j' does not visit the vertex v_{2i}, then there is $\gamma_l' \in \hat{\gamma}^c$ $(l \neq j)$ that visits v_{2i}. But, then e_{2i} would have to appear in γ_l' too, which contradicts (i). Hence, γ_j' must visit the vertex v_{2i}. Using the same arguments, one can show that γ_j' contains the vertices $v_{3i}, \ldots, v_{s+1i}$ and the edges $e_{3i}, \ldots, e_{s+1i}$. Therefore we obtain $\gamma_i^{cyc} = \gamma_j'$, since γ_i is a cycle.

By the symmetry of the arguments, the above shows that there is a one-to-one correspondence between the elements of γ^c and $\hat{\gamma}^c$ that do not contain edges e_1 and e_{m+1}. Further, γ^c has an element γ_1 that contains the edges e_1 and e_{m+1}, iff there is an element γ_1' in $\hat{\gamma}^c$ that contains both e_1 and e_{m+1}. Moreover, it must be the case that $\gamma_1 = \gamma_1'$ since both γ_1 and γ_1' contain the edges/vertices not visited by γ and the rest of the paths in γ^c and $\hat{\gamma}^c$. This proves that $\gamma^c = \hat{\gamma}^c$, i.e., γ^c is unique for a given Γ and a given Hamiltonian polygonal path γ in Γ. □

Lemma 6.12. *Let Γ be a simple assembly graph with no endpoints, and let γ be a Hamiltonian polygonal path, with γ^c its minimal complementary Hamiltonian set of polygonal paths. Then the number of connected components of $\tilde{\Gamma}_\gamma$ is $|\gamma^c|$ if γ is a cycle, and $|\gamma^c| + 1$, otherwise.*

Proof. Let Γ be a simple assembly graph, with a Hamiltonian polygonal path

$$\gamma = ((e_1 \setminus v_0), v_1, e_2, \ldots, v_{m-1}, e_m, v_m, (e_{m+1} \setminus v_{m+1})),$$

and let $\gamma^c = \{\gamma_1, \ldots, \gamma_s\}$ be a minimal complementary Hamiltonian set of polygonal paths to γ. Let $C = \{C_1, \ldots, C_p\}$ be the set of components of $\tilde{\Gamma}_\gamma$. As mentioned in Remark 5.2, each component in C contains edges and vertices inherited from Γ. Denote with $E(C_i)$ the set of edges inherited from Γ that are contained in C_i, and let $E' = \{E(C_1), \ldots, E(C_p)\}$. Note that $E(C_i)$ is non-empty for each $i \in \{1, 2, \ldots, p\}$, and, for each edge $e \in E(\Gamma)$, there is exactly one $i \in \{1, 2, \ldots, p\}$ such that $e \in E(C_i)$. Hence E' is a partition of $E(\Gamma)$.

First, assume that there is a path in γ^c that contains edges from γ. Then, by Lemma 6.11 (iv), such a path is unique in γ^c and contains both e_1 and e_{m+1}. Without loss of generality, let $\gamma_1 \in \gamma^c$ be such a path. Define

$$G_1 = \{e \in E(\Gamma) \mid \text{ the edge } e \text{ intersects with } \gamma \text{ or } \gamma_1\}, \quad \text{and}$$
$$G_i = \{e \in E(\Gamma) \mid \text{ the edge } e \text{ intersects with } \gamma_i\} \quad \text{for } i = 2, \ldots, s.$$

Note that if an edge $e \in E(\Gamma)$ intersects with γ_i, then it is contained in γ_i, or intersects at one of the ends of γ_i. By (i) and (ii) from Lemma 6.11 $G = \{G_1, \ldots, G_k\}$ is a partition on $E(V)$.

Next we show that $G = E'$. Let

$$\gamma_i = ((e_1^i \setminus v_0^i), v_1^i, e_2^i, \ldots, v_{m_i-1}^i, e_{m_i}^i, v_{m_i}^i, (e_{m_i+1}^i \setminus v_{m_i+1}^i)), \quad \text{for } i \in \{1, 2, \ldots, s\}.$$

By the definition of smoothing, all edges from γ appear in one component of C, and therefore in one of the set in E', say $E(C_1)$. Since $e_1^1 = e_{m+1} \in E(C_1)$, by the definition of vertex smoothing we have $e_2^1 \in E(C_1)$. Then $e_2^1 \in E(C_1)$ implies $e_3^1 \in E(C_1)$, and inductively, one obtains that every edge from γ_1 is in $E(C_1)$. For any fixed $i \in \{2, \ldots, s\}$, by Lemma 6.11 (iii) we have that $e_1^i = e_{m+1}^i$, i.e., γ_i is cyclic. If C' is a cyclic component in C with $e_1^i \in E(C')$, by the definition of smoothing, the edges e_j^i belong to the same partition set $E(C')$ from E' for all $j \in \{2, \ldots, m_i + 1\}$. Hence, for each $G_i \in G$, there is $E(C_j) \in E'$ such that $G_i \subset E(C_j)$. As both G and E' are partitions of $E(\Gamma)$, the previous statement implies $G_i = E(C_j)$. Hence, $G = E'$. The proof is similar if we assume that there is no element in γ^c that contains edges from γ. In that case, consider $G = \{G_i | i \in \{0, 1, 2, \ldots, s\}\}$, where $G_0 = \{e \in E(\Gamma) \mid \text{ the edge } e \text{ intersects with } \gamma\}$ and $G_i = \{e \in E(\Gamma) \mid \text{ the edge } e \text{ intersects with } \gamma_i\}$ for $i \in \{1, 2, \ldots, s\}$. □

Throughout the rest of the section we assume that Γ is a simple assembly graph with no endpoints. Let γ be a Hamiltonian polygonal path in Γ, and $\gamma^c = \{\gamma_1, \gamma_2, \ldots, \gamma_k\}$ be a minimal Hamiltonian set of polygonal paths complementary to γ. Denote by $\gamma^{c'}$ the subset of γ^c that consists of the paths that are edge disjoint from γ.

Let $S \subseteq V(\Gamma)$. For $\gamma_i \in \gamma^{c'}$, let C_i be the component of $\tilde{\Gamma}_{(\gamma,S)}$ that contains edges inherited from γ_i. When $S = V(\Gamma)$ such a component is connected in $\tilde{\Gamma}_{(\gamma,V(\Gamma))}$, by the proof of Lemma 6.12.

Lemma 6.13. *Let $\gamma_i \in \gamma^{c'}$ and $S_i = \{v | v \in V(\Gamma), v$ appears in $\gamma_i\}$. Then, for any successful set $S \subset V(\Gamma)$, C_i is a connected component of $\tilde{\Gamma}_{(\gamma,S)}$ if and only if $S_i \subset S$.*

Proof. First, assume that C_i is a connected component of $\tilde{\Gamma}_{(\gamma,S)}$. This implies that each edge of γ_i appears in C_i. That is possible only if every vertex of γ_i is smoothed. Therefore, $S_i \subset S$.

Conversely, let $S_i \subset S$. Since γ_i has no edges from γ, by Lemma 6.11(ii) it is cyclic. Each vertex that belongs to γ_i is smoothed. By smoothing the vertices from γ_i (i.e., S_i), one obtains a connected component that contains each edge from γ_i, as explained in the proof of Lemma 6.12. By definition, that component is C_i, so C_i is a component of $\tilde{\Gamma}_{(\gamma,S)}$. □

For the rest of the section, we call C_i the component of $\tilde{\Gamma}_{(\gamma,S)}$ that corresponds to S_i, if they are related as in Lemma 6.13. Keeping the same definitions of Γ, γ, γ^c, S_i, and C_i, we characterize the successful sets as follows.

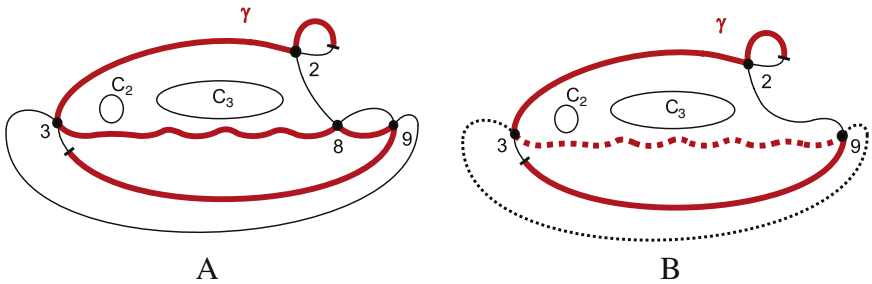

Fig. 16. Successful and unsuccessful smoothings. (A) Successful smoothing of Γ in Example 6.10 relative set $S = \{4, 5, 6, 7\} = \{4\} \cup \{5, 6, 7\} = S_1 \cup S_2$. The resulting graph contains γ in a single component. (B) Unsuccessful smoothing of Γ in Example 6.10 relative set $S = \{4, 5, 6, 7, 8\} = \{4\} \cup \{5, 6, 7\} \cup \{8\} = S_1 \cup S_2 \cup \{8\}$. The set $\{8\}$ is not linear in $\tilde{\Gamma}_{(\gamma, S_2 \cup S_3)} \setminus (C_2 \cup C_3)$ since the resulting graph contains γ on two distinct components, indicated by full and dotted lines.

Proposition 6.14. *A subset $S \subset V(\Gamma)$ is successful iff there is $I \subseteq \{1, \ldots, k\}$ such that S can be written as $S_I \cup S'$, where $S_I = \bigcup_{i \in I} S_i$, S' is linear in $\tilde{\Gamma}_{(\gamma, S_I)} \setminus (\bigcup_{i \in I} C_i)$.*

Proof. Let $S \subset V(\Gamma)$ be a successful set. If S is linear in Γ, then $\tilde{\Gamma}_\gamma$ is a single component and we can take $I = \emptyset$. So, $S = \emptyset \cup S$. If S is not linear, then $\tilde{\Gamma}_{(\gamma, S)}$ has more than one connected component. Let $C = \{C_1, \ldots, C_k\}$ be the set of all connected components of $\tilde{\Gamma}_{(\gamma, S)}$. One of them contains γ, since S is successful. Let $I = \{i \mid C_i$ is a component in $\tilde{\Gamma}_{(\gamma, S)}$ that does not contain $\gamma\}$. If C' is a component of $\tilde{\Gamma}_{(\gamma, S)}$ that does not contain γ, then the corresponding vertex set $S' = \{v | v \in V(\Gamma), v$ appears in $\gamma' \in \gamma^{c'}\}$ is a subset of S, by Lemma 6.13. Hence, $S_i \subset S$ for any S_i that corresponds to some connected component C_i such that $i \in I$. Therefore, $S_I \subset S$. Furthermore, because S is successful note that $\tilde{\Gamma}_{(\gamma, S)} \setminus \bigcup_{i \in I} C_i$ has one transverse arc component that contains γ. This implies that $S' = S \setminus S_I$ must be linear for $\tilde{\Gamma}_{(\gamma, S_I)} \setminus \bigcup_{i \in I} C_i$. We have $S = S_I \bigcup S'$.

Conversely, let $S = S_I \bigcup S'$ for some $S_I = \cup_{i \in I} S_i$, $I \subseteq \{1, \ldots, k\}$. Note that $\tilde{\Gamma}_{(\gamma, S_I)}$ contains the connected component C_i that corresponds to S_i for each $i \in I$, by Lemma 6.13. Since none of those components contains γ, $\tilde{\Gamma}_{(\gamma, S_I)} \setminus \bigcup_{i \in I} C_i$ is an assembly graph that contains γ. Since S' is linear in $\tilde{\Gamma}_{(\gamma, S_I)} \setminus \bigcup_{i \in I} C_i$, $\tilde{\Gamma}_{(\gamma, S_I \bigcup S')} \setminus \bigcup_{i \in I} C_i$ is a simple assembly graph that contains γ. Hence, $\tilde{\Gamma}_{(\gamma, S)}$ has a transverse arc component that contains γ, or, in other words, S is a successful set. □

Example 6.15. For the simple assembly graph Γ and the polygonal Hamiltonian path γ given in Example 6.10 (see Fig. 15), $\gamma^{c'} = \{\gamma_2, \gamma_3\}$. Furthermore, $S_2 = \{4\}$, $S_3 = \{5, 6, 7\}$ and let C_2 and C_3 be components of $\tilde{\Gamma}_\gamma$ that correspond to S_2 and S_3, respectively. Then, the set $\{2, 3, 4\}$ is successful by Proposition 6.14, since $\{2, 3, 4\} = S_2 \cup \{2, 3\}$ and $\{2, 3\}$ is linear in $\tilde{\Gamma}_{(\gamma, S_2)} \setminus C_2$ (see Fig. 14(B)). The set $\{4, 8\}$ is not successful by Example 6.7 (see Fig. 14(C)). It can be written as $S_2 \cup \{8\}$, but $\{8\}$ is not linear in $\tilde{\Gamma}_{(\gamma, S_2)} \setminus C_2$. The set $\{4, 5, 6, 7\}$ can be written as $S_2 \cup S_3 \bigcup \emptyset$. It is clear from Fig. 16(A) that $\emptyset$ is linear in $\tilde{\Gamma}_{(\gamma, S_2 \cup S_3)} \setminus (C_2 \cup C_3)$. Hence $\{4, 5, 6, 7\}$ is a successful set. On the other hand, the set $\{4, 5, 6, 7, 8\}$ is not successful, since $\{8\}$ is not linear in $\tilde{\Gamma}_{(\gamma, S_2 \cup S_3)} \setminus (C_2 \cup C_3)$ (see Fig. 16(B)).

Definition 6.16. Given a simple assembly graph Γ, let $\mathcal{S} = \{S_1, \ldots, S_k\}$ be an ordered partition of $V(G)$. Denote $S'_1 = \emptyset$ and $S'_i = \cup_{j=1}^{i-1} S_j$. We say that for a Hamiltonian polygonal path γ in Γ, $\mathcal{S}$ is a *successful smoothing strategy* for Γ with respect to γ if S_i is successful in $\tilde{\Gamma}_{(\gamma, S'_i)}$, for every $i = 1, \ldots, k$.

Example 6.17. For the simple assembly graph Γ and the polygonal Hamiltonian path γ given in Fig. 17(A), consider the sets $S_1 = \{2, 3\}$ and $S_2 = \{5, 6\}$. The smoothing $\tilde{\Gamma}_{(\gamma, S_1)}$ is depicted in Fig. 17(B), and since γ belongs to a single transverse component of $\tilde{\Gamma}_{(\gamma, S_1)}$, S_1 is successful. The set $S_2 = \{5, 6\}$ is not successful, as seen in $\tilde{\Gamma}_{(\gamma, S_2)}$ depicted in Fig. 17(C) since portions of γ belong to two transverse components. Furthermore, $S_1 \cup S_2 = V(\Gamma)$. Therefore, the sequence (S_1, S_2) is a successful smoothing strategy, but (S_2, S_1) is not a successful smoothing strategy for Γ with respect to γ.

Lemma 6.18. *Let Γ be a simple assembly graph, γ be a Hamiltonian polygonal path in Γ and let $S \subset S' \subset V(\Gamma)$. If both S and S' are successful with respect to γ, then $S_1 = S' - S$ is successful for $\tilde{\Gamma}_{(\gamma, S)}$.*

Proof. Let $S \subset S' \subset V(\Gamma)$, and suppose that both S and S' are successful. Then $\tilde{\Gamma}_{(\gamma, S)}$ is an assembly graph that has a transverse arc component containing γ. The application of S'-partial smoothing on $\tilde{\Gamma}_{(\gamma, S)}$ results in $\tilde{\Gamma}_{(\gamma, S')}$. However, $\tilde{\Gamma}_{(\gamma, S')}$ is an assembly graph that has a transverse component containing γ, since S' is successful in Γ. Therefore, $S_1 = S' - S$ is successful for $\tilde{\Gamma}_{(\gamma, S)}$. □

Proposition 6.19. *There is a one-to-one correspondence between the set of successful smoothing strategies and the set of all nested sequences: $P_1 \subset P_2 \subset \cdots \subset P_k = V(\Gamma)$, where P_i is successful for Γ for every $i = 1, \ldots, k$.*

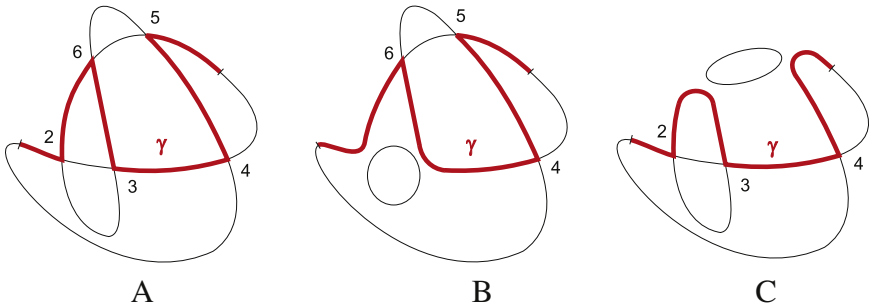

Fig. 17. (A) An example of an assembly graph and the corresponding Hamiltonian polygonal path. (B) S_1-smoothing of the graph in (A), for the successful set $S_1 = \{2, 3\}$. (C) S_2-smoothing of the graph in (A), for the set $S_2 = \{5, 6\}$. S_2 is not successful and therefore can not be the first smoothing set in a smoothing strategy.

Proof. Let $\{S_1, \ldots, S_k\}$ be a successful smoothing strategy for Γ with respect to γ. Then, S_i is successful in $\tilde{\Gamma}_{(\gamma, S_i')}$, where $S_i' = \cup_{h=1}^{i-1} S_h$. Hence, $P_i = S_i \cup S_i'$ is successful in Γ. Therefore, $P_1 \subset P_2 \subset \cdots \subset P_k$ is a nested sequence of successful sets in Γ.

Conversely, let $P_1 \subset P_2 \subset \cdots \subset P_k = V(\Gamma)$ be a nested sequence where P_i is successful for Γ for every $i = 1, \ldots, k$. Define $S_i = P_i - P_{i-1}$. Since $P_{i-1} \subset P_i$ and both P_i and P_{i-1} are successful sets in Γ, by Lemma 6.18, $P_i - P_{i-1}$ is successful in $\Gamma_{(\gamma, P_{i-1})}$. On the other hand, $P_{i-1} = \cup_{h=1}^{i-1} S_h = S_i'$. Therefore, $S_i = P_i - P_{i-1}$ is a successful set in $\Gamma_{(\gamma, S_i')}$. The ordered set $\{S_1, \ldots, S_k\}$ consists of pairwise disjoint subsets of $V(\Gamma)$ whose union is $V(\Gamma)$ and each S_i is successful in $\tilde{\Gamma}_{(\gamma, S_i')}$. Hence $S_1, \ldots, S_k$ is a successful smoothing strategy for γ. $\square$

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